

Trigonometry

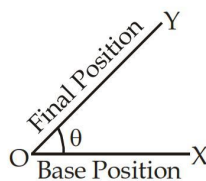
Trigonometry: Trigonometry word is derived from two greek words.

- Trigonon : Triangle
- Metron : Measure.

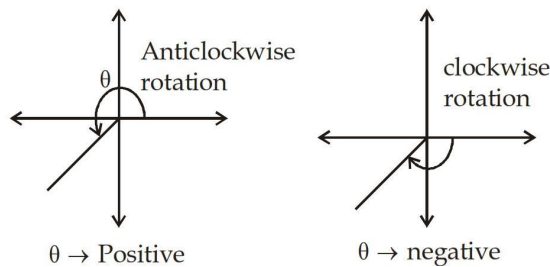
So, generally, it is a branch of mathematics that defines the relationship between the angle and the sides of the triangle.

Angle: Angle is defined by the particular rotation of line. From initial position to a terminal position.

Here, OX – Base position OY – Final Position θ – Angle.



Convention of sign for angle : Generally we consider anticlockwise rotation as positive and clockwise rotation as negative.



Measurement of angle: There are three different systems for angle measurement in trigonometry.

- Centesimal system
- Sexagesimal system
- Circular system

Centesimal system: One right angle is divided into 100 parts and each part is called a grade (1g)

$$90^\circ = 100g$$

$$21g = 100' \text{ (Centesimal minute)} \quad 1' = 100'' \text{ (Centesimal record)}$$

Sexagesimal: One right angle is divided into 60 equal parts and each part is called a degree (1°).

$$1^\circ = 60' \text{ (Sexagesimal minute)}$$

$$1' = 60'' \text{ (Sexagesimal second)}$$

Circular system: Unit of measurement of angle in this system is radian. It is denoted by 1^c .

Radian: It is the angle subtended at the centre by the arc of length equal to radius of circle.

- In circle, circumference = $\pi \cdot \text{diameter}$.

$$\text{Where } \pi = 3.1416 \text{ as } \frac{22}{7}$$

- $\pi \text{ radian} = 180^\circ$ (Relationship between degree and radian)
- When an arc subtends an angle θ radian at the centre of a circle of radius r then,

$$\theta = \frac{\text{arc}}{\text{radius}}$$

- $1 \text{ radian} = \frac{180}{\frac{22}{7}} = 57^\circ 16' 22''$

- Thus, to change degree into radian, multiply by $\frac{\pi}{180^\circ}$ and to change radian into degree multiply by $\frac{180^\circ}{\pi}$

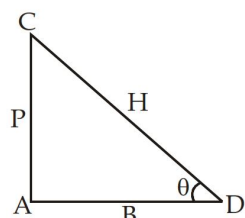
Example : Find radian measure of 120° .

Solution : $120^\circ = 120 \times \frac{\pi}{180^\circ} = \frac{2}{3} \pi \text{ radian}$

Example : Find the value of $\frac{3}{5}$ radian in degree.

Solution : $\frac{3}{5} \text{ rad} = \frac{3}{5} \times \frac{180}{\pi} = \frac{3}{5} \times \frac{7}{22} \times 180^\circ = \frac{378^\circ}{11} = 34^\circ 21' 49''$

Trigonometric Ratio :



Let $\angle ABC = \theta$ (Acute angle of right angle triangle)

Here, $AB = \text{Base (B)}$

$AC = \text{Perpendicular (P)}$

$BC = \text{Hypotenuse (H)}$

There are six trigonometric ratios.

- $\sin \theta = \frac{P}{H} = \frac{B}{H} \cdot \frac{P}{B}$ $\sin \theta = \frac{\text{Perpendicular}}{\text{Hypotenuse}}$

- $\cos \theta = \frac{B}{H} = \frac{P}{H} \cdot \frac{B}{P}$ $\cos \theta = \frac{\text{Base}}{\text{Hypotenuse}}$

- $\tan \theta = \frac{P}{B} = \frac{P}{H} \cdot \frac{H}{B}$ $\tan \theta = \frac{\text{Perpendicular}}{\text{Base}}$

- $\cot \theta = \frac{B}{P} = \frac{B}{H} \cdot \frac{H}{P}$ $\cot \theta = \frac{\text{Base}}{\text{Perpendicular}}$

- $\sec \theta = \frac{H}{B} = \frac{P}{H} \cdot \frac{H}{B}$ $\sec \theta = \frac{\text{Hypotenuse}}{\text{Base}}$

- $\csc \theta = \frac{H}{P} = \frac{P}{H} \cdot \frac{H}{P}$ $\csc \theta = \frac{\text{Hypotenuse}}{\text{Perpendicular}}$

$\frac{PBP}{HHB} \rightarrow \frac{\text{Pandit Badri Prasad}}{\text{Har Har Bole}} \text{ (To memorize)}$

Reciprocal relations

- $\sin \theta = \frac{1}{\csc \theta} \rightarrow \csc \theta = \frac{1}{\sin \theta}$

- $\cos \theta = \frac{1}{\sec \theta} \rightarrow \sec \theta = \frac{1}{\cos \theta}$

- $\tan \theta = \frac{1}{\cot \theta} \rightarrow \cot \theta = \frac{1}{\tan \theta}$

Multiple Relation

- $\sin \theta \cdot \csc \theta = 1$

- $\sec \theta \cdot \cos \theta = 1$

- $\tan \theta \cdot \cot \theta = 1$

Quotient relation:

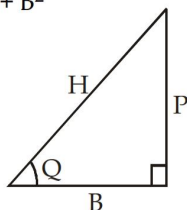
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$

- $\cot \theta = \frac{\cos \theta}{\sin \theta}$

Pythagoras Theorem:

- $(\text{Hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{Base})^2$

$$H^2 = P^2 + B^2$$



- The number which satisfy this relation are called Pythagoras triplets. All multiples and submultiple of equation also satisfy the relation.

Trigonometric ratio of sum specific Angles

Trigonometric Ratios	0°	30°	45°	60°	90°	180°	270°	360°
	0rad.	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
sin θ	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos θ	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1
tan θ	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞	0	∞	0
cot θ	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0	∞	0	∞
sec θ	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞	-1	∞	1
cosec θ	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1	∞	-1	∞

Some Important Result

- If $\sin x = 0$ or $\tan x = 0$, then $x = n\pi$

$$\text{If } \cos x = 0 \text{ or } \cot x = 0, \text{ then } x = (2n + 1) \frac{\pi}{2}$$

- $\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$

- $\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$

- $\sin 22\frac{1}{2}^\circ = \frac{\sqrt{2} - \sqrt{2}}{2}$

- $\cos 22\frac{1}{2}^\circ = \frac{\sqrt{2} + \sqrt{2}}{2}$

- $\tan 22\frac{1}{2}^\circ = \sqrt{2} - 1$

- $\cot 22\frac{1}{2}^\circ = \sqrt{2} + 1$

- $\sin 18^\circ = \cos 72^\circ = \frac{\sqrt{5} - 1}{4}$

- $\cos 18^\circ = \sin 72^\circ = \frac{\sqrt{10 + 2\sqrt{5}}}{4}$

- $\sin 36^\circ = \cos 54^\circ = \frac{\sqrt{10 - 2\sqrt{5}}}{4}$

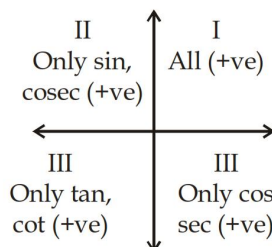
- $\cos 36^\circ = \sin 54^\circ = \frac{\sqrt{5} + 1}{4}$

Sign of Trigonometric functions in different Quadrant

We can obtain the sign of trigonometric function from Quadrant chart.

- We can remember as

Add Sugar To coffee
 ↓ ↓ ↓ ↓
 All sin/ cosec tan/ cot cos/ sec



Trigonometric ratio of Negative and Associated angle

Angle	$-\theta$	$(90 - \theta)$	$(90 + \theta)$	$(180 - \theta)$	$(180 + \theta)$	$(360 - \theta)$	$(360 + \theta)$
$\sin \theta$	$-\sin \theta$	$\cos \theta$	$\cos \theta$	$\sin \theta$	$-\sin \theta$	$-\sin \theta$	$\sin \theta$
$\cos \theta$	$\cos \theta$	$\sin \theta$	$-\sin \theta$	$-\cos \theta$	$-\cos \theta$	$\cos \theta$	$\cos \theta$
$\tan \theta$	$-\tan \theta$	$\cot \theta$	$-\cot \theta$	$-\tan \theta$	$\tan \theta$	$-\tan \theta$	$\tan \theta$

From the above table we observe that only trigonometric ratio of $(90 \pm \theta)$, and $(270 \pm \theta)$ change and they change as.

$$\sin \xrightarrow{\text{Change}} \cos$$

$$\tan \xrightarrow{\text{Change}} \cot$$

$$\text{cosec} \xrightarrow{\text{Change}} \sec$$

- Trigonometric ratio of $(180 \pm \theta)$ and $(360 \pm \theta)$ do not change
- Sign convention is used according to Quadrant.

Example :
$$= \frac{\sin 12^\circ}{\cos 78^\circ} = \frac{\sin (90^\circ - 78^\circ)}{\cos 78^\circ} = \frac{\cos 78^\circ}{\cos 78^\circ} = 1$$

Example : $\sin 2295^\circ$

$$\begin{array}{r} 6 \\ 360^\circ \overline{) 2295^\circ} \\ \underline{2160} \\ 135 \end{array} \quad \text{So, } \sin 2295^\circ = (360^\circ \times 6 + 135^\circ) = \sin 135^\circ$$

$$\text{Now } \sin (135^\circ) = \sin (90^\circ + 45^\circ) = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

Some results

when $A + B = 90^\circ$ i.e. sum of angles of acute angle triangle = 90°

- $\sin A = \cos B$
- $\tan A \cdot \tan B = 1$
- $\cot A \cdot \cot B = 1$ when $A + B = 90^\circ$

Basic Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta \Rightarrow \sec^2 \theta - \tan^2 \theta = 1$

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1, \quad (\sec \theta + \tan \theta) = \frac{1}{(\sec \theta - \tan \theta)}$$

$$\text{If } \sec \theta + \tan \theta = P, \quad \text{then, } \sec \theta - \tan \theta = \frac{1}{P}$$

- $1 + \cot^2 \theta = \text{cosec}^2 \theta$

$$\text{cosec}^2 \theta - \cot^2 \theta = 1, \quad (\text{cosec} \theta + \cot \theta) = \frac{1}{(\text{cosec} \theta - \cot \theta)}, \quad \text{If } \text{cosec} \theta + \cot \theta = P, \quad \text{then, } \text{cosec} \theta - \cot \theta = \frac{1}{P}$$

Advanced trigonometric Identities

1. $\sin(A + B) = \sin A \cos B + \cos A \sin B$
3. $\cos(A + B) = \cos A \cos B - \sin A \sin B$
5. $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
7. $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$
9. $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
11. $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
13. $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$
15. $\sin C + \sin D = 2 \sin \frac{C + D}{2} \cos \frac{C - D}{2}$
17. $\cos C + \cos D = 2 \cos \frac{C + D}{2} \cos \frac{C - D}{2}$
19. $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$
21. $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
22. $\sin 3A = 3 \sin A - 4 \sin^3 A$
24. $\tan 3A = \frac{3 \tan A - \tan^3 A}{1 - 3 \tan^2 A}$
26. $\cos A = \cos^2\left(\frac{A}{2}\right) - \sin^2\left(\frac{A}{2}\right)$ etc.
 $= 1 - 2 \sin^2 A/2$
 $= 2 \cos^2 A/2 - 1$.
28. $\tan(A + B + C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan A \tan C}$
29. $\sin A \sin 2A \cdot \sin 4A = \frac{1}{4} \sin 3A$
31. $\tan A \tan 2A \tan 4A = \tan 3A$
2. $\sin(A - B) = \sin A \cos B - \cos A \sin B$
4. $\cos(A - B) = \cos A \cos B + \sin A \sin B$
6. $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
8. $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
10. $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
12. $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
14. $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$
16. $\sin C - \sin D = 2 \cos \frac{C + D}{2} \sin \frac{C - D}{2}$
18. $\cos C - \cos D = 2 \sin \frac{C + D}{2} \sin \frac{D - C}{2}$
20. $\cos 2A = \cos^2 A - \sin^2 A = 1 - 2 \sin^2 A = 2 \cos^2 A - 1$
 $= \frac{1 - \tan^2 A}{1 + \tan^2 A}$
23. $\cos 3A = 4 \cos^3 A - 3 \cos A$
25. $\sin A = 2 \sin\left(\frac{A}{2}\right) \cos\left(\frac{A}{2}\right) = \frac{2 \tan(A/2)}{1 + \tan^2(A/2)}$
27. $\tan A = \frac{2 \tan(A/2)}{1 - \tan^2(A/2)}$
30. $\cos A \cos 2A \cdot \cos 4A = \frac{1}{4} \cos 3A$

Range of trigonometric functions

S.No.	Trigonometric Ratio	Range
1	$\sin A$	$[-1, 1]$
2	$\cos A$	$[-1, 1]$
3	$\tan A$	$[-\infty, \infty]$
4	$\cot A$	$[-\infty, \infty]$
5	$\sec A$	$(-\infty, -1] \cup [1, \infty)$
6	$\operatorname{cosec} A$	$(-\infty, -1] \cup [1, \infty)$

$\sin^2 A$ and $\cos^2 A$ has minimum value 0 and maximum value 1 so range is (0, 1)

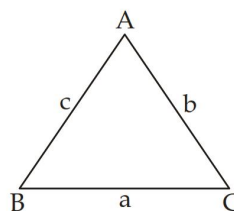
For values of $0^\circ \leq \theta \leq 90^\circ$ then minimum and maximum value are given as

S. No.	Trigonometric Functions	Minimum value	Maximum value
1	$a \sin \theta + b \cos \theta$	$-\sqrt{a^2 + b^2}$	$\sqrt{a^2 + b^2}$
2	$a \sin \theta + b \operatorname{cosec} \theta$	$2\sqrt{ab}$	∞
3	$a \cos \theta + b \sec \theta$	$2\sqrt{ab}$	∞
4	$a \tan \theta + b \cot \theta$	$2\sqrt{ab}$	∞
5	$a \sin^2 \theta + b \operatorname{cosec}^2 \theta$	$2\sqrt{ab}$	∞
6	$a \cos^2 \theta + b \sec^2 \theta$	$2\sqrt{ab}$	∞
7	$a \tan^2 \theta + b \cot^2 \theta$	$2\sqrt{ab}$	∞
8	$a \sin^2 \theta + b \cos^2 \theta$	a or b which ever is lowest	a or b which ever is greater
9	$a \sec \theta + b \operatorname{cosec} \theta$	$(\sqrt{a} + \sqrt{b})^2$	∞

Sine Rule:

In any $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \text{ where } R \text{ is circumradius.}$$



Cosine Rule:

$$a^2 = b^2 + c^2 - 2bc \cos A, \quad b^2 = a^2 + c^2 - 2ac \cos B, \quad c^2 = a^2 + b^2 - 2ab \cos C$$

Important Results:

- $\sin \theta + \cos \theta = a$ then, $\sin \theta - \cos \theta = \sqrt{2 - a^2}$
- $a \sin \theta + b \cos \theta = p$ and, $a \cos \theta - b \sin \theta = q$ then, $a^2 + b^2 = p^2 + q^2$, $q = \pm \sqrt{a^2 + b^2 - p^2}$
- $a \sin \theta + b \cos \theta = \sqrt{a^2 + b^2}$ then, $\sin \theta = \frac{a}{\sqrt{a^2 + b^2}}$, $\cos \theta = \frac{b}{\sqrt{a^2 + b^2}}$
- If $a \sec \theta - b \tan \theta = \sqrt{a^2 - b^2}$ then, $\sec \theta = \frac{a}{\sqrt{a^2 - b^2}}$, $\tan \theta = \frac{b}{\sqrt{a^2 - b^2}}$
- If $a \operatorname{cosec} \theta - b \cot \theta = \sqrt{a^2 - b^2}$ then, $\operatorname{cosec} \theta = \frac{a}{\sqrt{a^2 - b^2}}$, $\cot \theta = \frac{b}{\sqrt{a^2 - b^2}}$
- Some trigonometric function can be solved easily by considering it as algebraic function.

Example : $\sin \theta + \operatorname{cosec} \theta = 2$

It can be considered as $x + \frac{1}{x}$, where $x = \sin \theta$

$$\text{so, } x = 1 \text{ and } \frac{1}{x} = 1, \quad \sin \theta = 1 \text{ and } \frac{1}{\operatorname{cosec} \theta} = 1$$

Foundation

Questions

1. If $\sin \theta = -\frac{12}{13}$ and $\pi < \theta < \frac{3\pi}{2}$, then $\sec \theta$ is:
 - (a) $-\frac{13}{5}$
 - (b) $-\frac{12}{5}$
 - (c) $-\frac{5}{12}$
 - (d) $\frac{13}{5}$
2. $\sin \theta = \frac{5}{13}$, then the values of $\tan \theta$ and $\sec \theta$ are respectively.
 - (a) $\frac{12}{13}$ and $\frac{13}{12}$
 - (b) $\frac{5}{12}$ and $\frac{13}{12}$
 - (c) $\frac{13}{12}$ and $\frac{5}{12}$
 - (d) $\frac{12}{5}$ and $\frac{13}{5}$
3. If $\sin \theta = \frac{5}{13}$ and θ is acute, then the value of $\sqrt{(1 + \tan \theta)(1 - \tan \theta)}$ is:
 - (a) $\frac{13}{5}$
 - (b) $\frac{12}{15}$
 - (c) $\frac{13}{12}$
 - (d) None of these
4. If $\tan \theta = \left(\frac{3}{4}\right)$ and $0 < \theta < 90^\circ$, then the value of $(\sin \theta \cos \theta)$ is:
 - (a) $\frac{3}{5}$
 - (b) $\frac{4}{5}$
 - (c) $\frac{12}{25}$
 - (d) None of these
5. If $\tan \theta = \left(\frac{3}{4}\right)$, then the value of $(\cos^2 \theta - \sin^2 \theta)$ is:
 - (a) $\frac{1}{4}$
 - (b) $\frac{1}{2}$
 - (c) $\frac{4}{25}$
 - (d) $\frac{7}{25}$
6. If $\cot^2 \theta = \left(\frac{7}{8}\right)$ and $0 < \theta < 90^\circ$, then the value of $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$ is:
 - (a) $\frac{7}{8}$
 - (b) $\frac{7}{6}$
 - (c) $\frac{7}{5}$
 - (d) $\frac{7}{4}$
7. $(\sin A + \cos A)^2 + (\sin A - \cos A)^2$ is equal to:
 - (a) 1
 - (b) 2
 - (c) $3 \cos A$
 - (d) $4 \sin A$
8. $\sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}}$ is equal to:
 - (a) $\sec x + \tan x$
 - (b) $\sec x - \tan x$
 - (c) $\operatorname{cosec} x + \cot x$
 - (d) $\operatorname{cosec} x - \cot x$
9. $(1 - \tan \theta)^2 + (1 + \tan \theta)^2$ is equal to:
 - (a) $2 \tan^2 \theta$
 - (b) $2 \tan \theta$
 - (c) $4 \tan \theta$
 - (d) $2 \sec^2 \theta$
10. $(\cos^4 \theta - \sin^4 \theta)$ is equal to:
 - (a) $2 \sin^2 \theta - 1$
 - (b) $1 - 2 \cos^2 \theta$
 - (c) $\sin^2 \theta - \cos^2 \theta$
 - (d) None of these
11. $\frac{\sin \theta}{(1 + \cos \theta)}$ is equal to:
 - (a) $\frac{\sin \theta}{\cos \theta}$
 - (b) $\frac{\cos \theta - 1}{\sin \theta}$
 - (c) $\frac{1 - \cos \theta}{\sin \theta}$
 - (d) $\frac{\sin \theta + 1}{\cos \theta}$
12. If $0 \leq \theta \leq 90^\circ$, then: $\left(\frac{5 \cos \theta - 4}{3 - 5 \sin \theta} - \frac{3 + 5 \sin \theta}{4 + 5 \cos \theta}\right)$ is equal to:
 - (a) 0
 - (b) 1
 - (c) $\frac{1}{2}$
 - (d) $\frac{1}{4}$
13. $(\cot \theta \cdot \operatorname{cosec} \theta \cdot \sin \theta \cdot \tan \theta)$ is equal to:
 - (a) $2 \cos \theta$
 - (b) $3 \sin \theta$
 - (c) 1
 - (d) $\sin \theta$
14. $\left(\frac{\tan A}{\sec A - 1} + \frac{\tan A}{\sec A + 1}\right)$ is equal to:
 - (a) $2 \tan A$
 - (b) $2 \tan A \sec A$
 - (c) $2 \operatorname{cosec} A$
 - (d) $2 \sec A$
15. $\left(\frac{\sin A + \sin B}{\cos A + \cos B} + \frac{\cos A - \cos B}{\sin A - \sin B}\right)$ is equal to:
 - (a) $\tan A \tan B$
 - (b) $\tan A + \tan B$
 - (c) 1
 - (d) 0
16. If a and b are real numbers such that $a \cos \theta + b \sin \theta = 4$ and $a \sin \theta - b \cos \theta = 3$, then $(a^2 + b^2)$ is:
 - (a) 7
 - (b) 12
 - (c) 25
 - (d) $\sqrt{12}$

17. The value of $(\sec^2 60^\circ - 1)$ is:
 (a) 2 (b) 3
 (c) 4 (d) 5
18. $(\sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ)$ is equal to:
 (a) 0 (b) $\frac{1}{4}$
 (c) 4 (d) 1
19. If $x \tan 45^\circ \cos 60^\circ = \sin 60^\circ \cot 60^\circ$, then x is equal to:
 (a) 1 (b) $\sqrt{3}$
 (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{2}}$
20. The value of $\left(\sin \frac{\pi}{6} + \cos \frac{\pi}{3} + \tan \frac{\pi}{4}\right)$ is:
 (a) 1 (b) 0
 (c) -1 (d) None of these
21. If p is the value of $p = \sin^2 20^\circ + \sin^2 40^\circ + \sin^2 50^\circ + \sin^2 70^\circ$, then:
 (a) $0 < p < 1$ (b) $1 < p < 1.5$
 (c) $1.5 < p < 2$ (d) $p = 2$
22. The value of $(\sin^2 15^\circ + \sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ + \sin^2 75^\circ)$ is:
 (a) 1 (b) 1.5
 (c) 2.5 (d) 3
23. If $\frac{x \operatorname{cosec}^2 30^\circ \sec^2 45^\circ}{8 \cos^2 45^\circ \sin^2 60^\circ} = \tan^2 60^\circ - \tan^2 30^\circ$, then the value of x is:
 (a) 1 (b) -1
 (c) 2 (d) 0
24. The value of $(\cos^2 15^\circ + \cos^2 75^\circ)$ is:
 (a) $\sin^2 15^\circ + \sin^2 75^\circ$ (b) $\tan^2 15^\circ + \tan^2 75^\circ$
 (c) $\cot^2 15^\circ + \cot^2 75^\circ$ (d) $\sec^2 15^\circ + \sec^2 75^\circ$
25. If A and B are supplementary angles, then:
 (a) $\sin A = \sin B$ (b) $\cos A = \cos B$
 (c) $\tan A = \tan B$ (d) $\sec A = \operatorname{cosec} B$
26. If $\tan A = \frac{1}{2}$ and $\tan B = \frac{1}{3}$, then $(A + B)$ is:
 (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$
27. If $\tan x = \frac{m}{m+1}$ and $\tan y = \frac{1}{(2m+1)}$, then $(x + y)$ is:
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
 (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
28. $\sin(45^\circ + \theta) - \sin(45^\circ - \theta)$ is equal to:
 (a) $\sin \theta$ (b) $\cos \theta$
 (c) $\sqrt{2} \cos \theta$ (d) $\sqrt{2} \sin \theta$
29. If $\sin 2x = \frac{1}{5}$, the value of $(\sin x + \cos x)$ is:
 (a) $\sqrt{\frac{7}{5}}$ (b) $\sqrt{\frac{6}{5}}$
 (c) $\sqrt{\frac{4}{5}}$ (d) $\sqrt{\frac{2}{5}}$
30. The maximum and minimum values of $(1 + \cos 2x)$ are:
 (a) -1 and 1 (b) 1 and 2
 (c) $-\frac{1}{2}$ and $\frac{1}{2}$ (d) 0 and 2
31. If $\sin x = \frac{1}{2}$, then $\tan 2x$ equals:
 (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{1}{\sqrt{3}}$
 (c) $\frac{\sqrt{3}}{2}$ (d) $\sqrt{3}$
32. If $\sin \theta + \cos \theta = \sqrt{2}$, then the value of θ is:
 (a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{6}$ (d) $\frac{\pi}{2}$
33. The values of θ satisfying the equation $2 \sin^2 \theta = 3 \sin \theta - 1$ are:
 (a) $\frac{\pi}{3}, \frac{\pi}{4}$ (b) $\frac{\pi}{4}, \frac{\pi}{6}$
 (c) $\frac{\pi}{4}, \frac{\pi}{2}$ (d) $\frac{\pi}{2}, \frac{\pi}{6}$
34. $(1 - \sin^2 \alpha) \tan^2 \alpha$ is equal to:
 (a) $\cos^2 \alpha$ (b) $\tan^2 \alpha$
 (c) $\sin^2 \alpha$ (d) $\cos \alpha$
35. $\sin^2 \theta = \frac{p^2 + q^2}{2pq}$, if
 (a) $p = q$ (b) $p = -q$
 (c) $2p = 3q$ (d) $p = 2q$
36. Given that $\operatorname{cosec} \theta = \frac{5}{4}$, if θ is in the second quadrant, then the value of $\cos \theta$ will be:
 (a) $\frac{4}{5}$ (b) $-\frac{5}{3}$
 (c) $-\frac{3}{5}$ (d) $\frac{3}{5}$

37. If $\sin \alpha + \cos \beta = 2$ ($0^\circ \leq \beta < \alpha \leq 90^\circ$), then $\sin\left(\frac{2\alpha + \beta}{3}\right) =$
- (a) $\sin \frac{\alpha}{2}$ (b) $\cos \frac{\alpha}{3}$ (c) $\sin \frac{\alpha}{3}$ (d) $\cos \frac{2\alpha}{3}$
38. If $\cos^4 \theta - \sin^4 \theta = \frac{2}{3}$, then the value of $2 \cos^2 \theta - 1$ is:
- (a) 0 (b) 1 (c) $\frac{2}{3}$ (d) $\frac{3}{2}$
39. $\frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta}$ is equal to:
- (a) $2 \sec^2 \theta$ (b) $2 \operatorname{cosec}^2 \theta$ (c) $1^2 - \sin^2 \theta$ (d) $2 \cos^2 \theta$
40. $\cos^2 \theta (1 + \tan^2 \theta)$ is equal to:
- (a) 1 (b) $\frac{1}{2}$ (c) 2 (d) 3
41. If $\sec^2 \theta - \tan^2 \theta = 1$, then what will be the value of θ :
- (a) 30° (b) 60° (c) 90° (d) All of them
42. If $\tan x = \frac{4}{3}$, then the value of: $\sqrt{\frac{(1 - \sin x)(1 + \sin x)}{(1 + \cos x)(1 - \cos x)}}$
- (a) $\frac{9}{16}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{16}{9}$
43. If $\sec^2 \theta + \tan^2 \theta = \frac{7}{12}$, then find the value of $\sec^4 \theta - \tan^4 \theta$:
- (a) $\frac{7}{12}$ (b) $\frac{1}{2}$ (c) $\frac{5}{12}$ (d) 1
44. If $\sin^2 60^\circ + \cos^2 (3x - 9^\circ) = 1$ then value of x is:
- (a) 24° (b) 23° (c) 22° (d) 21°
45. If $\sin(x - y) = \frac{1}{2}$ and $\cos(x + y) = \frac{1}{2}$ then value of x and y :
- (a) $15^\circ, 45^\circ$ (b) $45^\circ, 15^\circ$ (c) $30^\circ, 60^\circ$ (d) $60^\circ, 30^\circ$
46. What is the value of $(\sec^2 \theta - 1)(\operatorname{cosec}^2 \theta - 1)$?
- (a) 0 (b) 1 (c) 2 (d) -1
47. If $2 \cos 3\theta_1 = 1$ and $2 \sin 2\theta_2 = \sqrt{3}$ then what will be the value of θ_1 and θ_2 ?
- (a) $30^\circ, 20^\circ$ (b) $60^\circ, 40^\circ$ (c) $20^\circ, 30^\circ$ (d) $45^\circ, 45^\circ$
48. If $\tan(\alpha + \beta) = 1$ and $\tan(\alpha - \beta) = \frac{1}{\sqrt{3}}$ then find the value of $\tan(2\alpha + 2\beta)$:
- (a) Positive Integer (b) Negative Integer (c) Zero (d) Infinite
49. The minimum value of $\sin \theta + \cos \theta$ is:
- (a) -1 (b) 0 (c) $\sqrt{2}$ (d) $-\sqrt{2}$
50. $\sin \alpha = \sin \beta$ and $\cos \alpha = \cos \beta$ then $\sin\left(\frac{\alpha - \beta}{\sqrt{2}}\right)$ is equal to:
- (a) $\frac{3}{2}$ (b) $\frac{5}{2}$ (c) 0 (d) $< \frac{3}{2}$

Moderate

1. If $\tan \theta = \frac{x}{y}$, then the value of $\left(\frac{x \sin \theta + y \cos \theta}{x \sin \theta - y \cos \theta}\right)$ equals:
- (a) $\frac{x^2 + y^2}{x^2 - y^2}$ (b) $\frac{x^2 - y^2}{x^2 + y^2}$ (c) $\frac{x}{\sqrt{x^2 + y^2}}$ (d) $\frac{y}{\sqrt{x^2 + y^2}}$
2. If $7 \sin^2 \theta + 3 \cos^2 \theta = 4$ and $0 < \theta < \frac{\pi}{2}$, then the value of $\tan \theta$ is:
- (a) $\frac{4}{7}$ (b) $\frac{3}{7}$ (c) $\frac{2}{7}$ (d) $\frac{1}{7}$
3. If $\sec^2 A = 3$, $0 < A < \frac{\pi}{2}$, then the value of $\frac{\tan^2 A - \operatorname{cosec}^2 A}{\tan^2 A + \operatorname{cosec}^2 A}$ is:
- (a) $\frac{4}{7}$ (b) $\frac{3}{7}$ (c) $\frac{2}{7}$ (d) $\frac{1}{7}$

4. $\sqrt{\frac{1+\sin A}{1-\sin A}}$ is equal to:
- (a) $\sec A + \tan A$ (b) $\sec^2 A + \tan^2 A$
(c) $\sec^2 A - \tan^2 A$ (d) $\sec A \tan A$
5. $(\sec^4 A - \sec^2 A)$ is equal to:
- (a) $\tan^2 A + \tan^4 A$ (b) $\tan^4 A - \tan^2 A$
(c) $\tan^2 A - \tan^4 A$ (d) None of these
6. The value of $(\cot^4 \theta - \operatorname{cosec}^4 \theta + \cot^2 \theta + \operatorname{cosec}^2 \theta)$ is:
- (a) 1 (b) 0
(c) -1 (d) 2
7. $\left(\frac{\sin \theta}{1-\cot \theta} + \frac{\cos \theta}{1-\tan \theta}\right)$ is equal to:
- (a) $\cos \theta + \sin \theta$ (b) 0
(c) 1 (d) $\tan \theta$
8. $\left(\frac{\cot \theta}{\cot \theta - \cot 3\theta} + \frac{\tan \theta}{\tan \theta - \tan 3\theta}\right)$ is equal to:
- (a) 0 (b) 1
(c) -1 (d) 2
9. If $\sin \theta + \cos \theta = m$ and $\sin^3 \theta + \cos^3 \theta = n$, then
- (a) $m^3 - 3m + n = 0$ (b) $n^3 - 3n + 2m = 0$
(c) $m^3 - 3m + 2n = 0$ (d) $m^3 + 3m + 2n = 0$
10. The value of $(\cot 15^\circ \cot 16^\circ \cot 17^\circ \dots \cot 73^\circ \cot 74^\circ \cot 75^\circ)$ is:
- (a) $\frac{1}{2}$ (b) 0
(c) 1 (d) -1
11. The value of $(\sin 79^\circ \cos 11^\circ + \sin 11^\circ \cos 79^\circ)$ is:
- (a) 1 (b) 0
(c) $\frac{1}{\sqrt{2}}$ (d) $\frac{1}{2}$
12. The value of $\cos 20^\circ + \cos 40^\circ + \cos 60^\circ + \dots \cos 160^\circ + \cos 180^\circ$ is:
- (a) 0 (b) 1
(c) -1 (d) $\frac{1}{2}$
13. $\tan 22^\circ 30'$ is equal to:
- (a) $\sqrt{2} - 1$ (b) $\sqrt{2} + 1$
(c) $\sqrt{2}$ (d) $\frac{1}{2}$
14. The value of $4 \sin 45^\circ \cos 15^\circ$ is:
- (a) $\sqrt{3} + 1$ (b) $\sqrt{3} - 1$
(c) 1 (d) 2
15. If $\sin \theta = \frac{1}{2}$, $\cos \phi = 1$, where $0 < \theta < \frac{\pi}{2}$ and $0 < \phi \leq \frac{\pi}{2}$, then $\cot(\theta + \phi)$ is equal to:
- (a) $\frac{\sqrt{3}}{2}$ (b) $\sqrt{3}$
(c) 1 (d) 2
16. If ABCD is a cyclic quadrilateral, then:
- (a) $\cos A - \cos B + \cos C + \cos D = 0$
(b) $\cos A + \cos B + \cos C - \cos D = 0$
(c) $\cos A + \cos B - \cos C - \cos D = 0$
(d) $\cos A - \cos B + \cos C - \cos D = 0$
17. The value of $\frac{1 + \tan 15^\circ}{1 - \tan 15^\circ}$ is:
- (a) $-\frac{2}{\sqrt{3}}$ (b) $\frac{1}{2}$
(c) $\sqrt{3}$ (d) 2
18. If $0^\circ < A < 90^\circ$ and $\cos A - \sin A > 0$, then $\cos A + \sin A$ cannot be greater than:
- (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
(c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$
19. $\frac{\cot A + \operatorname{cosec} A - 1}{\cot A - \operatorname{cosec} A + 1}$ is equal to:
- (a) $\operatorname{cosec} A + \cot A$ (b) $\sec A + \cot A$
(c) $\operatorname{cosec} A + \tan A$ (d) $\operatorname{cosec} A - \cot A$
20. $\cos 3\theta + \sin 3\theta$ is maximum when θ is:
- (a) 15° (b) 30°
(c) 45° (d) 60°
21. The value of $\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ}$ is -
- (a) $\tan 56^\circ$ (b) $\sin 56^\circ$
(c) $\cos 56^\circ$ (d) $\cot 56^\circ$
22. $\sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right)$ is equal to:
- (a) $2 \sin \frac{\pi}{8}$ (b) $\sin A \sin \frac{\pi}{8}$
(c) $\frac{1}{2} \sin A$ (d) $\frac{1}{\sqrt{2}} \sin A$
23. Consider the following statements:
- $\sin \theta$ is an increasing function of θ in the first quadrant.
 - $\cos \theta$ is an increasing function of θ in the first and second quadrant.
 - $0 < \sin \theta < \cos \theta$ for $0 < \theta < \frac{\pi}{4}$
 - $\sin \theta < \cos \theta < 0$ for $\frac{\pi}{2} < \theta < \frac{3\pi}{4}$

Of these statements,

- (a) 1 and 3 are correct (b) 2 and 4 are correct
(c) 1, 2 and 4 are correct (d) 1, 3 and 4 are correct

24. If $\tan x = \frac{1}{\sqrt{3}}$ and $\sin y = \frac{1}{\sqrt{2}}$, then the value of $(x + y)$

is:

- (a) Zero (b) more than 90°
(c) between 0° and 90° (d) between 45° and 135°

25. The value of $\cot 10^\circ \cdot \cot 20^\circ \cdot \cot 60^\circ \cdot \cot 70^\circ \cdot \cot 80^\circ$ is:

- (a) 1 (b) -1

- (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

26. If $\sin \alpha \sec (30^\circ + \alpha) = 1$ ($0 < \alpha < 60^\circ$), then the value of $\sin \alpha + \cos 2\alpha$ is:

- (a) 1 (b) $\frac{2 + \sqrt{3}}{2\sqrt{3}}$

- (c) 0 (d) $\sqrt{2}$

27. If θ be a positive acute angle satisfying $\cos^2 \theta + \cos^4 \theta = 1$, then the value of $\tan^2 \theta + \tan^4 \theta$ is:

- (a) $\frac{3}{2}$ (b) 1

- (c) $\frac{1}{2}$ (d) 0

28. The value of $\tan 4^\circ \cdot \tan 43^\circ \cdot \tan 47^\circ \cdot \tan 86^\circ$ is:

- (a) 0 (b) 1

- (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

29. If $\sin 17^\circ = \frac{x}{y}$, then the value of $\sec 17^\circ - \sin 73^\circ$ is:

- (a) $\frac{y^2 - x^2}{xy}$ (b) $\frac{x^2}{\sqrt{y^2 - x^2}}$

- (c) $\frac{x^2}{y\sqrt{y^2 + x^2}}$ (d) $\frac{x^2}{y\sqrt{y^2 - x^2}}$

30. If $0 < x < \frac{\pi}{2}$ and $\sec x = \operatorname{cosec} y$, then the value of $\sin (x + y)$ is:

- (a) 0 (b) 1

- (c) $\frac{1}{2}$ (d) $\frac{1}{\sqrt{3}}$

31. The value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$ is:

- (a) 1 (b) 0

- (c) $\sqrt{3}$ (d) $\frac{1}{\sqrt{3}}$

32. If A, B and C be the angles of a triangle, then of the following the incorrect relation is:

(a) $\sin\left(\frac{A+B}{2}\right) = \cos \frac{C}{2}$ (b) $\cos\left(\frac{A+B}{2}\right) = \sin \frac{C}{2}$

(c) $\tan\left(\frac{A+B}{2}\right) = \sec \frac{C}{2}$ (d) $\cot\left(\frac{A+B}{2}\right) = \tan \frac{C}{2}$

33. The simplified value of $(\sec A - \cos A)^2 + (\operatorname{cosec} A - \sin A)^2 - (\cot A - \tan A)^2$ is:

- (a) 0 (b) $\frac{1}{2}$

- (c) 1 (d) 2

34. If θ be an acute angle and $7 \sin^2 \theta + 3 \cos^2 \theta = 4$, then the value of $\operatorname{cosec} \theta$ is:

- (a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$

- (c) 1 (d) 2

35. The numerical value of

$\cot 18^\circ \left(\cot 72^\circ \cos^2 22^\circ + \frac{1}{\tan 72^\circ \sec^2 68^\circ} \right)$ is:

- (a) 1 (b) $\sqrt{2}$

- (c) 3 (d) $\frac{1}{\sqrt{3}}$

36. If $\operatorname{cosec} 39^\circ = x$, the value of

$\frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ - \frac{1}{\sin^2 51^\circ \sec^2 39^\circ}$ is:

- (a) $\sqrt{x^2 - 1}$ (b) $\sqrt{1 - x^2}$

- (c) $x^2 - 1$ (d) $1 - x^2$

37. If $\frac{\tan \theta + \cot \theta}{\tan \theta - \cot \theta} = 2$, ($0 \leq \theta \leq 90^\circ$), then the value of $\sin \theta$ is:

- (a) $\frac{2}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$

- (c) $\frac{1}{2}$ (d) 1

38. If $\sin \theta + \operatorname{cosec} \theta = 2$, then the value of $\sin^5 \theta + \operatorname{cosec}^5 \theta$ when $0^\circ \leq \theta \leq 90^\circ$ is:

- (a) 0 (b) 1

- (c) 10 (d) 2

39. $\cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 85^\circ + \cos^2 90^\circ$ is equal to:

- (a) $7\frac{1}{2}$ (b) $8\frac{1}{2}$

- (c) 9 (d) $9\frac{1}{2}$

40. If $2\cos\theta - \sin\theta = \frac{1}{\sqrt{2}}$, ($0^\circ < \theta < 90^\circ$) the value of $2\sin\theta + \cos\theta$ is:
- (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$
(c) $\frac{3}{\sqrt{2}}$ (d) $\frac{\sqrt{2}}{3}$
41. If $\frac{\sin\theta + \cos\theta}{\sin\theta - \cos\theta} = 3$, then the value of $\sin^4\theta - \cos^4\theta$ is:
- (a) $\frac{1}{5}$ (b) $\frac{2}{5}$
(c) $\frac{3}{5}$ (d) $\frac{4}{5}$
42. The value of $\frac{\sin 39^\circ}{\cos 51^\circ} + 2 \tan 11^\circ \tan 31^\circ \tan 45^\circ \tan 59^\circ \tan 79^\circ - 3(\sin^2 21^\circ + \sin^2 69^\circ)$ is:
- (a) 2 (b) -1
(c) 1 (d) 0
43. If $A = \tan 11^\circ \tan 29^\circ$, $B = 2 \cot 61^\circ \cot 79^\circ$, then:
- (a) $A = 2B$ (b) $A = -2B$
(c) $2A = B$ (d) $2A = -B$
44. If $\cos^2\alpha + \cos^2\beta = 2$, then the value of $\tan^3\alpha + \sin^5\beta$ is:
- (a) -1 (b) 0
(c) 1 (d) $\frac{1}{\sqrt{3}}$
45. If $\sin\theta - \cos\theta = \frac{7}{13}$ and $0 < \theta < 90^\circ$, then the value of $\sin\theta + \cos\theta$ is:
- (a) $\frac{17}{13}$ (b) $\frac{13}{17}$
(c) $\frac{1}{13}$ (d) $\frac{1}{17}$
46. If x, y are acute angles, $0 < x + y < 90^\circ$ and $\sin(2x - 20^\circ) = \cos(2y + 20^\circ)$, then the value of $\tan(x + y)$ is:
- (a) $\frac{1}{\sqrt{3}}$ (b) $\frac{\sqrt{3}}{2}$
(c) $\sqrt{3}$ (d) 1
47. The simplified value of $(\sec x \sec y + \tan x \tan y)^2 - (\sec x \tan y + \tan x \sec y)^2$ is:
- (a) -1 (b) 0
(c) $\sec^2 x$ (d) 1
48. $\sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ = ?$
- (a) $39\frac{1}{2}$ (b) $40\frac{1}{2}$
(c) 40 (d) $39\frac{1}{\sqrt{2}}$
49. If $\sin\theta + \operatorname{cosec}\theta = 2$, then value of $\sin^{100}\theta + \operatorname{cosec}^{100}\theta$ is equal to:
- (a) 1 (b) 2
(c) 3 (d) 100
50. $\tan 7^\circ \tan 23^\circ \tan 60^\circ \tan 67^\circ \tan 83^\circ$ is equal to:
- (a) 0 (b) $\sqrt{3}$
(c) $\frac{1}{\sqrt{3}}$ (d) 1

Difficult

1. The value of $\cot 82\frac{1}{2}^\circ$ is:
- (a) $\sqrt{6} + \sqrt{2} - 2 - \sqrt{3}$ (b) $\sqrt{6} - \sqrt{2} + 2 - \sqrt{3}$
(c) $\sqrt{6} + \sqrt{2} - 2 + \sqrt{3}$ (d) $\sqrt{6} - \sqrt{2} - 2 + \sqrt{3}$
2. The value of $\tan 142\frac{1}{2}^\circ$ is:
- (a) $\sqrt{2} + 2 - \sqrt{3} - \sqrt{6}$ (b) $\sqrt{2} - 2 + \sqrt{3} - \sqrt{6}$
(c) $\sqrt{2} + 2 - \sqrt{3} + \sqrt{6}$ (d) $\sqrt{2} - 2 - \sqrt{3} + \sqrt{6}$
3. If $\alpha = \frac{\pi}{13}$, $\cos\alpha \cdot \cos 2\alpha \cdot \cos 3\alpha \cdot \cos 4\alpha \cdot \cos 5\alpha \cdot \cos 6\alpha =$
- (a) $\frac{1}{16}$ (b) 16
(c) $\frac{1}{64}$ (d) 64
4. $\sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \cdot \sin \frac{7\pi}{14} \cdot \sin \frac{9\pi}{14} \cdot \sin \frac{11\pi}{14} \cdot \sin \frac{13\pi}{14} =$
- (a) $\frac{1}{64}$ (b) 8
(c) $\frac{1}{8}$ (d) 64
5. $\sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{5\pi}{8} + \sin^4 \frac{7\pi}{8} =$
- (a) $\frac{5}{2}$ (b) $\frac{1}{2}$
(c) $\frac{3}{2}$ (d) 1
6. If $\sec\theta = x + \frac{1}{4x}$ ($0^\circ < \theta < 90^\circ$), then $\sec\theta + \tan\theta$ is equal to?

- (a) x (b) $\frac{1}{2x}$
- (c) $\frac{x}{2}$ (d) $2x$
7. If $(a^2 - b^2) \sin \theta + 2ab + \cos \theta = a^2 + b^2$, then find the value of $\tan \theta$
- (a) $\frac{1}{2ab}(a^2 + b^2)$ (b) $\frac{1}{2}(a^2 - b^2)$
- (c) $\frac{1}{2ab}(a^2 - b^2)$ (d) $\frac{1}{2}(a^2 + b^2)$
8. If $2 - \cos^2 \theta = 3 \sin \theta \cdot \cos \theta$, $\sin \theta \neq \cos \theta$, then find the value of $\tan \theta$
- (a) $\frac{1}{2}$ (b) 0
- (c) $\frac{2}{3}$ (d) $\frac{1}{3}$
9. $\sin^3 \theta + \sin^3 (120^\circ + \theta) + \sin^3 (240^\circ + \theta) =$
- (a) $\frac{3}{4} \sin 3\theta$ (b) $\frac{3}{4} \sin\left(\frac{\theta}{3}\right)$
- (c) $-\frac{3}{4} \sin 3\theta$ (d) 0
10. $\cot \theta + \cot (60^\circ + \theta) + \cot (120^\circ + \theta) =$
- (a) $\frac{1}{3} \cot 3\theta$ (b) $3 \cot\left(\frac{\theta}{3}\right)$
- (c) $\frac{1}{3} \cot\left(\frac{\theta}{3}\right)$ (d) $3 \cot(3\theta)$
11. $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha =$
- (a) $\cot \alpha$ (b) $\tan \alpha$
- (c) $\cot(2\alpha)$ (d) $\tan 2\alpha$
12. $\frac{1}{\cos 290^\circ} + \frac{1}{\sqrt{3} \sin 250^\circ}$ is equal to :
- (a) $\frac{\sqrt{3}}{4}$ (b) $\frac{4}{\sqrt{3}}$
- (c) $\frac{2}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{2}$
13. $(\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ)$ is equal to:
- (a) 2 (b) 3
- (c) 4 (d) None of these
14. Suppose α and β are such that $\pi < (\alpha - \beta) < 3\pi$. If $\sin \alpha + \sin \beta = -\frac{21}{65}$ and $\cos \alpha + \cos \beta = -\frac{27}{65}$ then the value of $\cos\left(\frac{\alpha - \beta}{2}\right)$ is:
- (a) $-\frac{3}{\sqrt{130}}$ (b) $\frac{3}{\sqrt{130}}$
- (c) $\frac{6}{65}$ (d) $-\frac{6}{65}$
15. If $\sin (10^\circ 6' 32'') = a$, then find the value of $\cos (79^\circ 53' 28'') + \tan (10^\circ 6' 32'')$
- (a) $\frac{\sqrt{1-a^2} + a}{\sqrt{1-a^2}}$ (b) $\frac{\sqrt{1-a^2} + 1}{\sqrt{1-a^2}}$
- (c) $\frac{a(\sqrt{1-a^2} + 1)}{\sqrt{1-a^2}}$ (d) $\frac{1 + \sqrt{1-a^2}}{\sqrt{1-a^2}}$
16. If $5 \cos \theta + 12 \sin \theta = 13$, then the value of $\operatorname{cosec} \theta$ is:
- (a) $\frac{5}{13}$ (b) $\frac{12}{5}$
- (c) $\frac{13}{12}$ (d) $\frac{5}{12}$
17. $\frac{\sec 29^\circ}{\operatorname{cosec} 61^\circ} + 2 \cot 8^\circ \cdot \cot 17^\circ \cdot \cot 45^\circ \cdot \cot 73^\circ \cdot \cot 82^\circ$
 $- 3(\sin^2 38^\circ + \sin^2 52^\circ) + 4\{\cos \theta \cdot \sin(90^\circ - \theta) + \sin \theta \cdot \cos(90^\circ - \theta)\} =$
- (a) 4 (b) 2
- (c) $\frac{1}{2}$ (d) None of these
18. $\sin(1 + \tan \theta) + \cos \theta(1 + \cos \theta) =$
- (a) $\tan \theta + \cot \theta$ (b) $\sec \theta + \operatorname{cosec} \theta$
- (c) $\tan \theta - \cot \theta$ (d) $\sec \theta - \operatorname{cosec} \theta$
19. Find the value of $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1}$ (where $\theta \neq \frac{\pi}{2}$)
- (a) $\frac{1 + \sin \theta}{\cos \theta}$ (b) $\frac{1 - \sin \theta}{\cos \theta}$
- (c) $\frac{1 - \cos \theta}{\sin \theta}$ (d) $\frac{1 + \cos \theta}{\sin \theta}$
20. $\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1} =$
- (a) $\frac{\sec \theta}{1 - \sin \theta}$ (b) $\frac{\cos \theta}{1 + \sin \theta}$
- (c) $\frac{\cos \theta}{1 - \sin \theta}$ (d) $\frac{\sec \theta}{1 + \sin \theta}$

21. If $\cot \theta + \cos \theta = m$ and $\cot \theta - \cos \theta = n$ then,

$$\frac{m^2 - n^2}{4\sqrt{mn}} =$$

- (a) -1 (b) 2
(c) $\frac{1}{2}$ (d) 1

22. If $\sec \theta + \tan \theta = p$, then the value of $\sin \theta$ is:

- (a) $\frac{1-p^2}{1+p^2}$ (b) $\frac{p^2}{1-p^2}$
(c) $\frac{1+p^2}{1-p^2}$ (d) $\frac{p^2}{1+p^2}$

23.
$$\frac{\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} - \sec \theta}{\sec \theta - \sqrt{\frac{1-\sin \theta}{1+\sin \theta}}} =$$

- (a) $\tan \theta$ (b) 1
(c) $\cot \theta$ (d) -1

24. If $(1 + \sin \alpha)(1 + \sin \beta)(1 + \sin \gamma) = (1 - \sin \alpha)(1 - \sin \beta)(1 - \sin \gamma)$, then each side is equal to ?

- (a) $\pm \cos \alpha \cos \beta \cos \gamma$ (b) $\pm \sin \alpha \sin \beta \sin \gamma$
(c) $\pm \sin \alpha \cos \beta \cos \gamma$ (d) $\pm \sin \alpha \sin \beta \cos \gamma$

25. Find the value of $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$

- (a) 1 (b) 2
(c) 0 (d) -1

26. If $\cos(\alpha - \theta) = a$, $\cos(\beta - \theta) = b$, then the value of $\sin^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$ is:

- (a) $a^2 + b^2$ (b) $a^2 - b^2$
(c) $b^2 - a^2$ (d) $-a^2 - b^2$

27. Find the value of

$$\cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8}.$$

- (a) 1 (b) -1
(c) 0 (d) $\frac{3}{2}$

28.
$$\frac{2}{\sqrt{2+\sqrt{2+\sqrt{2+2\cos 4x}}}} = ?$$

- (a) $\sec x$ (b) $\sec \frac{x}{2}$
(c) $\operatorname{cosec} x$ (d) 1

29.
$$\left(1 + \cos \frac{\pi}{8}\right)\left(1 + \cos \frac{3\pi}{8}\right)\left(1 + \cos \frac{5\pi}{8}\right)\left(1 + \cos \frac{7\pi}{8}\right) = ?$$

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{1}{8}$ (d) $\frac{1}{16}$

30. $(\sec x \cdot \sec y + \tan x \cdot \tan y)^2 - (\sec x \cdot \tan y + \tan x \cdot \sec y)^2$ in its simplest form, is:

- (a) -1 (b) 0
(c) $\sec^2 x$ (d) 1

Previous Year (Memory Based)

1. The value of $\frac{\sin A}{1 + \cos A} + \frac{\sin A}{1 - \cos A}$ is ($0^\circ < A < 90^\circ$):

- (a) $2 \operatorname{cosec} A$ (b) $2 \sec A$
(c) $2 \sin A$ (d) $2 \cos A$

2. If $r \sin \theta = 1$, $r \cos \theta = \sqrt{3}$, then the value of $(\sqrt{3} \tan \theta + 1)$ is:

- (a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$
(c) 1 (d) 2

3. The value of $\sin^2 65^\circ + \sin^2 25^\circ + \cos^2 35^\circ + \cos^2 55^\circ$ is:

- (a) 0 (b) 1
(c) 2 (d) $\frac{1}{2}$

4. If $x \sin 60^\circ \cdot \tan 30^\circ = \sec 60^\circ \cdot \cot 45^\circ$, then the value of x is:

- (a) 2 (b) $2\sqrt{3}$
(c) 4 (d) $4\sqrt{3}$

5. If $\theta = 60^\circ$, then $\frac{1}{2}\sqrt{1 + \sin \theta} + \frac{1}{2}\sqrt{1 - \sin \theta}$ is equal to:

- (a) $\cot \frac{\theta}{2}$ (b) $\sec \frac{\theta}{2}$
(c) $\sin \frac{\theta}{2}$ (d) $\cos \frac{\theta}{2}$

6. If $\frac{2 \tan^2 30^\circ}{1 - \tan^2 30^\circ} + \sec^2 45^\circ - \sec^2 0^\circ = x \sec 60^\circ$, then the value of x is:

- (a) 2 (b) 1
(c) 0 (d) -1

7. If $\tan \theta = \frac{\sin \alpha - \cos \alpha}{\sin \alpha + \cos \alpha}$, then $\sin \alpha + \cos \alpha$ is:

- (a) $\pm\sqrt{2}\sin\theta$ (b) $\pm\sqrt{2}\cos\theta$
 (c) $\pm\frac{1}{\sqrt{2}}\sin\theta$ (d) $\pm\frac{1}{\sqrt{2}}\cos\theta$
8. If $7\sin^2\theta + 3\cos^2\theta = 4$, ($0^\circ < \theta < 90^\circ$), then the value of $\tan^2\theta$ is:
 (a) $\frac{1}{3}$ (b) $\frac{1}{2}$
 (c) 1 (d) $\sqrt{3}$
9. If $\tan 9^\circ = \frac{p}{q}$, then the value of $\frac{\sec^2 81^\circ}{1 + \cot^2 81^\circ}$ is:
 (a) $\frac{q}{p}$ (b) 1
 (c) $\frac{p^2}{q^2}$ (d) $\frac{q^2}{p^2}$
10. If $\sec\theta + \tan\theta = 5$, then the value of $\frac{\tan\theta + 1}{\tan\theta - 1}$ is:
 (a) $\frac{11}{7}$ (b) $\frac{13}{7}$
 (c) $\frac{15}{7}$ (d) $\frac{17}{7}$
11. If $\tan^2\theta = 1 - e^2$, then the value of $\sec\theta + \tan^3\theta \operatorname{cosec}\theta$ is:
 (a) $(2 + e^2)^{\frac{3}{2}}$ (b) $(2 - e^2)^{\frac{1}{2}}$
 (c) $(2 + e^2)^{\frac{1}{2}}$ (d) $(2 - e^2)^{\frac{3}{2}}$
12. Find the minimum value $2\sin^2\theta + 3\cos^2\theta$?
 (a) 2 (b) -2
 (c) 6 (d) -4
13. If $x \sin 60^\circ \tan 30^\circ - \tan^2 45^\circ = \operatorname{cosec} 60^\circ \cot 30^\circ - \sec^2 45^\circ$, then $x =$
 (a) 2 (b) -2
 (c) 6 (d) -4
14. If $x = a \sec\alpha \cos\beta$, $y = b \sec\alpha \sin\beta$, $z = c \tan\alpha$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$ is:
 (a) 2 (b) 0
 (c) 1 (d) -1
15. If $\frac{\cos\alpha}{\cos\beta} = a$ and $\frac{\sin\alpha}{\sin\beta} = b$, then the value of $\sin^2\beta$ in terms of a and b is:
 (a) $\frac{a^2 + 1}{a^2 - b^2}$ (b) $\frac{a^2 - b^2}{a^2 + b^2}$
 (c) $\frac{a^2 - 1}{a^2 - b^2}$ (d) $\frac{a^2 - 1}{a^2 + b^2}$
16. The value of $\frac{\cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$ is:
 (a) $\frac{64}{\sqrt{3}}$ (b) $\frac{55}{12}$
 (c) $\frac{67}{12}$ (d) $\frac{67}{10}$
17. The value of $\sin^2 30^\circ \cos^2 45^\circ + 5 \tan^2 30^\circ + \frac{3}{2} \sin^2 90^\circ - 3 \cos^2 90^\circ$ is:
 (a) $3\frac{7}{24}$ (b) $3\frac{3}{24}$
 (c) $3\frac{1}{24}$ (d) $3\frac{5}{24}$
18. If $\cos^2\theta - \sin^2\theta = \frac{1}{3}$, where $0 \leq \theta \leq \frac{\pi}{2}$, then the value of $\cos^4\theta - \sin^4\theta$ is:
 (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
 (c) $\frac{1}{9}$ (d) $\frac{2}{9}$
19. If $\tan\theta = \frac{1}{\sqrt{11}}$ and $0 < \theta < \frac{\pi}{2}$, then the value of $\frac{\operatorname{cosec}^2\theta - \sec^2\theta}{\operatorname{cosec}^2\theta + \sec^2\theta}$ is:
 (a) $\frac{3}{4}$ (b) $\frac{4}{5}$
 (c) $\frac{5}{6}$ (d) $\frac{6}{7}$
20. The value of $\frac{1}{\sqrt{2}} \sin \frac{\pi}{6} \cos \frac{\pi}{4} - \cot \frac{\pi}{3} \sec \frac{\pi}{6} + \frac{5 \tan \frac{\pi}{4}}{12 \sin \frac{\pi}{2}}$ is equal to:
 (a) 0 (b) 1
 (c) 2 (d) $\frac{3}{2}$
21. If $\sin\theta = \frac{3}{5}$, then the value of $\frac{\tan\theta + \cos\theta}{\cot\theta + \operatorname{cosec}\theta}$ is equal to:
 (a) $\frac{29}{60}$ (b) $\frac{31}{60}$
 (c) $\frac{34}{60}$ (d) $\frac{37}{60}$

22. If $a \cos \theta + b \sin \theta = p$ and $a \sin \theta - b \cos \theta = q$, then the relation between a, b, p and q is:
 (a) $a^2 - b^2 = p^2 - q^2$ (b) $a^2 + b^2 = p^2 + q^2$
 (c) $a + b = p + q$ (d) $a - b = p - q$
23. If $\sin \theta + \cos \theta = \frac{7}{5}$, then the value of $\sin \theta \cdot \cos \theta$ is -
 (a) $\frac{4}{5}$ (b) $\frac{7}{8}$
 (c) $\frac{13}{12}$ (d) $\frac{12}{25}$
24. If $\sin 21^\circ = \frac{x}{y}$, then $\sec 21^\circ - \sin 69^\circ$ is equal to:
 (a) $\frac{x^2}{y\sqrt{y^2 - x^2}}$ (b) $\frac{y^2}{x\sqrt{y^2 - x^2}}$
 (c) $\frac{x^2}{y\sqrt{x^2 - y^2}}$ (d) $\frac{y^2}{x\sqrt{x^2 - y^2}}$
25. If $\sec \alpha + \tan \alpha = 2$, then the value of $\sin \alpha$ is (assume that $0 < \alpha < 90^\circ$)
 (a) 0.4 (b) 0.5
 (c) 0.6 (d) 0.8
26. If $3 \sin \theta + 5 \cos \theta = 5$, then the value of $5 \sin \theta - 3 \cos \theta$ will be:
 (a) ± 3 (b) ± 5
 (c) ± 2 (d) ± 1
27. If θ is an acute angle and $\tan \theta + \cot \theta = 2$, then the value of $\tan^5 \theta + \cot^5 \theta$ is:
 (a) 1 (b) 2
 (c) 3 (d) 4
28. The simple value of $\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots \tan 89^\circ$ is:
 (a) $\frac{1}{2}$ (b) 0
 (c) 1 (d) $\frac{2}{3}$
29. If $x \sin^2 60^\circ - \frac{3}{2} \sec 60^\circ \tan^2 30^\circ + \frac{4}{5} \sin^2 45^\circ \tan^2 60^\circ = 0$ then x is:
 (a) $-\frac{1}{15}$ (b) -4
 (c) $-\frac{4}{15}$ (d) -2
30. If $7 \sin \alpha = 24 \cos \alpha$; $0 < \alpha < \frac{\pi}{2}$, then the value of $14 \tan \alpha - 75 \cos \alpha - 7 \sec \alpha$ is equal to:
 (a) 3 (b) 4
 (c) 1 (d) 2
31. The value of x which satisfies the equation $2 \operatorname{cosec}^2 30^\circ + x \sin^2 60^\circ - \frac{3}{4} \tan^2 30^\circ = 10$ is:
 (a) 2 (b) 3
 (c) 0 (d) 1
32. If $2 \sin \theta + \cos \theta = \frac{7}{3}$ then the value of $(\tan^2 \theta - \sec^2 \theta)$ is:
 (a) 0 (b) -1
 (c) $\frac{3}{7}$ (d) $\frac{7}{3}$
33. If $29 \tan \theta = 31$, then the value of $\frac{1 + 2 \sin \theta \cos \theta}{1 - 2 \sin \theta \cos \theta}$ is equal to:
 (a) 810 (b) 900
 (c) 540 (d) 490
34. ABCD is a square of which AC is a diagonal. The value of $(\tan^2 \angle CAD + 1) \sin^2 \angle BAC$ is:
 (a) 2 (b) $\frac{1}{4}$
 (c) 1 (d) 0
35. If $\tan x = \sin 45^\circ \cdot \cos 45^\circ + \sin 30^\circ$ then the value of x is:
 (a) 30° (b) 45°
 (c) 60° (d) 90°
36. For any real values of θ , $\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} = ?$
 (a) $\cot \theta - \operatorname{cosec} \theta$ (b) $\sec \theta - \tan \theta$
 (c) $\operatorname{cosec} \theta - \cot \theta$ (d) $\tan \theta - \sec \theta$
37. If the sum and difference of two angles are 135° and $\frac{\pi}{12}$ respectively, then the value of the angles in degree measure are
 (a) $70^\circ, 65^\circ$ (b) $75^\circ, 60^\circ$
 (c) $45^\circ, 90^\circ$ (d) $80^\circ, 55^\circ$
38. In a $\triangle ABC$, $\angle B = \frac{\pi}{3}$, $\angle C = \frac{\pi}{4}$ and D divides BC internally in the ratio 1 : 3 then $\frac{\sin \angle BAD}{\sin \angle CAD}$ is equal to:
 (a) $\frac{1}{\sqrt{2}}$ (b) $\frac{1}{\sqrt{3}}$
 (c) $\frac{1}{\sqrt{6}}$ (d) $\sqrt{6}$
39. If $\sin 3A = \cos (A - 26^\circ)$, where $3A$ is an acute angle then the value of A is
 (a) 29° (b) 26°
 (c) 23° (d) 28°
40. Value of $\sec^2 \theta - \frac{\sin^2 \theta - 2 \sin^4 \theta}{2 \cos^4 \theta - \cos^2 \theta}$ is:
 (a) 1 (b) 2
 (c) -1 (d) 0

Foundation

Solutions

1. (a) Clearly, θ lies in 3rd quadrant and in this quadrant $\cos \theta$ is negative

$$\cos \theta = -\sqrt{1 - \sin^2 \theta} = -\sqrt{1 - \frac{144}{169}} = -\sqrt{\frac{25}{169}} = -\frac{5}{13}$$

$$\text{Hence, } \sec \theta = -\frac{13}{5}$$

2. (b) $\sin \theta = \frac{5}{13}$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{5}{12} \text{ and } \sec \theta = \frac{1}{\cos \theta} = \frac{13}{12}$$

3. (d) $\sin \theta = \frac{5}{13}$

$$\Rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\Rightarrow \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{5}{12} = \sqrt{(1 + \tan \theta)(1 - \tan \theta)}$$

$$= \sqrt{\frac{119}{144}} = \frac{\sqrt{119}}{12}$$

4. (c) $1 + \tan^2 \theta = \sec^2 \theta$

$$\Rightarrow \sec \theta = +\sqrt{1 + \tan^2 \theta} \quad [0^\circ < \theta < 90^\circ]$$

$$\Rightarrow \sec \theta = \sqrt{1 + \frac{9}{16}} = \sqrt{\frac{25}{16}} = +\frac{5}{4}$$

$$\Rightarrow \cos \theta = \frac{4}{5}$$

$$\therefore \sin \theta = \sqrt{1 - \cos^2 \theta} = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

$$\sin \theta \cdot \cos \theta = \frac{3}{5} \times \frac{4}{5} = \frac{12}{25}$$

5. (d) $\tan \theta = \frac{3}{4}$

$$\sec \theta = \sqrt{1 + \tan^2 \theta} = \sqrt{1 + \frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{5}{4}$$

$$\Rightarrow \cos \theta = \frac{4}{5}$$

$$\Rightarrow \sin \theta = \sqrt{1 - \frac{16}{25}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$$

$$\cos^2 \theta - \sin^2 \theta = \frac{16}{25} - \frac{9}{25} = \frac{7}{25}$$

$$6. (a); \frac{1 - \sin^2 \theta}{1 - \cos^2 \theta} = \frac{\cos^2 \theta}{\sin^2 \theta} = \cot^2 \theta = \frac{7}{8}$$

$$7. (b); (\sin A + \cos A)^2 + (\sin A - \cos A)^2$$

$$= \sin^2 A + \cos^2 A + 2 \sin A \cos A + \sin^2 A + \cos^2 A - 2 \sin A \cos A$$

$$= 2(\sin^2 A + \cos^2 A) = 2$$

$$8. (b) \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} = \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x} \times \frac{\sec x - \tan x}{\sec x - \tan x}}$$

$$= \sqrt{\frac{(\sec x - \tan x)^2}{\sec^2 x - \tan^2 x}}$$

$$= \sqrt{(\sec x - \tan x)^2} = \sec x - \tan x$$

$$9. (d) (1 - \tan \theta)^2 + (1 + \tan \theta)^2$$

$$= 1 + \tan^2 \theta - 2 \tan \theta + 1 + \tan^2 \theta + 2 \tan \theta$$

$$= 2(1 + \tan^2 \theta) = 2 \sec^2 \theta$$

$$10. (d) \cos^4 \theta - \sin^4 \theta = (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta)$$

$$= (\cos^2 \theta - \sin^2 \theta)$$

$$11. (c); \frac{\sin \theta(1 - \cos \theta)}{\sin^2 \theta} = \frac{1 - \cos \theta}{\sin \theta}$$

$$12. (a); = \frac{(5 \cos \theta - 4)(4 + 5 \cos \theta) - (3 + 5 \sin \theta)(3 - 5 \sin \theta)}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{[20 \cos \theta + 25 \cos^2 \theta - 16 - 20 \cos \theta]}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{-[9 - 15 \sin \theta + 15 \sin \theta - 25 \sin^2 \theta]}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{[20 \cos \theta + 25 \cos^2 \theta - 16 - 20 \cos \theta]}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)}$$

$$= \frac{-9 + 15 \sin \theta - 15 \sin \theta + 25 \sin^2 \theta}{(3 - 5 \sin \theta)(4 + 5 \cos \theta)} = 0$$

$$13. (c); \cot \theta \cdot \operatorname{cosec} \theta \cdot \sin \theta \cdot \tan \theta$$

$$= \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} \cdot \sin \theta \cdot \frac{\sin \theta}{\cos \theta} = 1$$

$$14. (c); \left(\frac{\tan A}{\sec A - 1} + \frac{\tan A}{\sec A + 1} \right)$$

$$= \frac{\tan A(\sec A + 1) + \tan A(\sec A - 1)}{(\sec A - 1)(\sec A + 1)}$$

$$= \frac{\tan A \sec A + \tan A + \tan A \sec A - \tan A}{\sec^2 A - 1}$$

$$= \frac{2 \tan A \sec A}{\tan^2 A} = \frac{2 \sec A}{\tan A} = 2 \frac{1}{\cos A} \times \frac{\cos A}{\sin A}$$

$$= 2 \operatorname{cosec} A$$

$$15. (d) \frac{\sin A + \sin B}{\cos A + \cos B} + \frac{\cos A - \cos B}{\sin A - \sin B}$$

$$= \frac{\sin^2 A - \sin^2 B + \cos^2 A - \cos^2 B}{(\cos A + \cos B)(\sin A - \sin B)}$$

$$= \frac{(\sin^2 A + \cos^2 A) - (\sin^2 B + \cos^2 B)}{(\cos A + \cos B)(\sin A - \sin B)} = 0$$

$$16. (c) a \cos \theta + b \sin \theta = 4, a \sin \theta - b \cos \theta = 3.$$

Squaring and adding, we get

$$a^2 \cos^2 \theta + b^2 \sin^2 \theta + a^2 \sin^2 \theta + b^2 \cos^2 \theta = 4^2 + 3^2 = 25$$

$$a^2(\sin^2 \theta + \cos^2 \theta) + b^2(\sin^2 \theta + \cos^2 \theta) = 25$$

$$\Rightarrow a^2 + b^2 = 25$$

$$17. (b); \sec^2 60^\circ - 1 = 2^2 - 1 = 3$$

$$18. (b); \sin^2 30^\circ + 4 \cot^2 45^\circ - \sec^2 60^\circ$$

$$= \left(\frac{1}{2} \right)^2 + 4 \times 1 - 2^2 = \frac{1}{4} + 4 - 4 = \frac{1}{4}$$

$$19. (a); x \tan 45^\circ \cos 60^\circ = \sin 60^\circ \cot 60^\circ$$

$$x \times 1 \times \frac{1}{2} = \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{3}}, x = 1$$

$$20. (d); \sin \frac{\pi}{6} + \cos \frac{\pi}{3} + \tan \frac{\pi}{4} = \frac{1}{2} + \frac{1}{2} + 1 = 2$$

$$\begin{aligned} 21. (d); \sin^2 20^\circ + \sin^2 40^\circ + \sin^2 50^\circ + \sin^2 70^\circ \\ = \sin^2 20^\circ + \sin^2 70^\circ + \sin^2 40^\circ + \sin^2 50^\circ \\ = \sin^2 20^\circ + \sin^2 (90^\circ - 20^\circ) + \sin^2 40^\circ + \sin^2 (90^\circ - 40^\circ) \\ = (\sin^2 20^\circ + \cos^2 20^\circ) + (\sin^2 40^\circ + \cos^2 40^\circ) \\ = 1 + 1 = 2 = p. \end{aligned}$$

$$\begin{aligned} 22. (c) \sin^2 15^\circ + \sin^2 30^\circ + \sin^2 45^\circ + \sin^2 60^\circ + \sin^2 75^\circ \\ = \sin^2 15^\circ + \sin^2 75^\circ + \sin^2 30^\circ + \sin^2 60^\circ + \sin^2 45^\circ \\ = (\sin^2 15^\circ + \cos^2 15^\circ) + (\sin^2 30^\circ + \cos^2 30^\circ) + \sin^2 45^\circ \\ = 1 + 1 + \frac{1}{2} = \frac{5}{2} = 2.5 \end{aligned}$$

$$23. (a) \frac{x \operatorname{cosec}^2 30^\circ \sec^2 45^\circ}{8 \cos^2 45^\circ \sin^2 60^\circ} = \tan^2 60^\circ - \tan^2 30^\circ$$

$$= \frac{x \times 2^2 \times (\sqrt{2})^2}{8 \times \left(\frac{1}{\sqrt{2}} \right)^2 \times \left(\frac{\sqrt{3}}{2} \right)^2} = (\sqrt{3})^2 - \left(\frac{1}{\sqrt{3}} \right)^2$$

$$= \frac{8x}{8 \times \frac{1}{2} \times \frac{3}{4}} = 3 - \frac{1}{3}, \frac{8x}{3} = \frac{8}{3}$$

$$\Rightarrow x = 1$$

$$24. (a); \cos^2 15^\circ + \cos^2 75^\circ \\ \cos^2 (90^\circ - 75^\circ) + \cos^2 (90^\circ - 15^\circ) \\ \sin^2 75^\circ + \sin^2 15^\circ$$

$$25. (a); A + B = 180^\circ$$

$$\Rightarrow A = 180^\circ - B$$

$$\Rightarrow \sin A = \sin (180^\circ - B) = \sin B$$

$$26. (c); \tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan (A + B) = \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{5}{6}}{1 - \frac{1}{6}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

$$\tan (A + B) = 1$$

$$\Rightarrow A + B = \frac{\pi}{4}.$$

$$27. (b); \tan (x + y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$= \frac{\frac{m}{m+1} + \frac{1}{2m+1}}{1 - \frac{m}{m+1} \cdot \frac{1}{2m+1}} = \frac{m(2m+1) + (m+1)}{(m+1)(2m+1) - m}$$

$$= \frac{2m^2 + 2m + 1}{2m^2 + 2m + 1} = 1, \tan (x + y) = 1$$

$$\Rightarrow x + y = \frac{\pi}{4}$$

28. (d) $\sin(45^\circ + \theta) - \sin(45^\circ - \theta)$

$$\begin{aligned} & \sin 45^\circ \cos \theta + \cos 45^\circ \sin \theta - [\sin 45^\circ \cos \theta - \cos 45^\circ \sin \theta] \\ & = \sin 45^\circ \cos \theta + \cos 45^\circ \sin \theta - \sin 45^\circ \cos \theta + \cos 45^\circ \sin \theta \end{aligned}$$

$$= 2 \cos 45^\circ \sin \theta = 2 \times \frac{1}{\sqrt{2}} \cdot \sin \theta = \sqrt{2} \sin \theta.$$

29. (b) $\sin 2x = \frac{1}{5}$ (given)

$$(\sin x + \cos x)^2 = \sin^2 x + \cos^2 x + 2 \sin x \cdot \cos x = 1 + \sin 2x$$

$$\Rightarrow (\sin x + \cos x)^2 = 1 + \frac{1}{5} = \frac{6}{5}$$

$$\Rightarrow \sin x + \cos x = \sqrt{\frac{6}{5}}$$

30. (d) Maximum value of $\cos 2x = 1$

Minimum value of $\cos 2x = -1$

$$\Rightarrow 0 \leq (1 + \cos^2 x) \leq 2$$

31. (d) $\sin x = \frac{1}{2}$

$$\Rightarrow \cos x = \sqrt{1 - \sin^2 x} = \sqrt{1 - \left(\frac{1}{2}\right)^2} = \sqrt{1 - \frac{1}{4}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \tan x = \frac{1}{\sqrt{3}}, \tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$= \frac{2 \times \frac{1}{\sqrt{3}}}{1 - \frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \sqrt{3}$$

32. (a) $\sin \theta + \cos \theta = \sqrt{2}$

$$\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta = 2$$

$$1 + \sin 2\theta = 2, \sin 2\theta = 1$$

$$\Rightarrow 2\theta = 90^\circ$$

$$\Rightarrow \theta = 45^\circ = \frac{\pi}{4}$$

33. (d) $2 \sin^2 \theta = 3 \sin \theta - 1$

$$2 \sin^2 \theta - 3 \sin \theta + 1 = 0$$

$$2 \sin^2 \theta - 2 \sin \theta - \sin \theta + 1 = 0$$

$$2 \sin \theta (\sin \theta - 1) - 1 (\sin \theta - 1) = 0$$

$$\Rightarrow (\sin \theta - 1) (2 \sin \theta - 1) = 0$$

$$\Rightarrow \sin \theta = 1 \text{ or } \sin \theta = \frac{1}{2}, \theta = \frac{\pi}{2}, \frac{\pi}{6}$$

34. (c) $(1 - \sin^2 \alpha) (\tan^2 \alpha) = \cos^2 \alpha \cdot \frac{\sin^2 \alpha}{\cos^2 \alpha} = \sin^2 \alpha.$

35. (a) When $p = q$,

$$\sin^2 \theta = \frac{2q^2}{2q^2} = 1, \text{ which is possible}$$

$$\text{When, } p = -q, \sin^2 \theta = \frac{2q^2}{-2q^2}$$

$$= -1, \text{ which is not possible}$$

$$\text{When, } 2p = 3q, p = \frac{3q}{2}$$

$$\sin^2 \theta = \frac{\frac{9}{4}q^2 + q^2}{2 \times \frac{3q}{2} \cdot q} = \frac{\frac{13q^2}{4}}{3q^2} = \frac{13}{12},$$

$$\text{which is not possible, when, } p = 2q,$$

$$\sin^2 \theta = \frac{4q^2 + q^2}{2 \cdot 2q \cdot q} = \frac{5q^2}{4q^2}$$

$$\Rightarrow \sin^2 \theta = \frac{5}{4}, \text{ which is not possible}$$

$$\text{So, correct option is (a).}$$

36. (c) $\sin \theta = \frac{4}{5}$

$$\cos \theta = \sqrt{1 - \sin^2 \theta} = -\sqrt{1 - \sin^2 \theta}$$

$$(\text{because } \theta \text{ is in second quadrant})$$

$$= -\sqrt{1 - \frac{16}{25}} = -\sqrt{\frac{9}{25}} = -\frac{3}{5}$$

37. (b) $\sin \alpha + \cos \beta = 2$

$$\text{maximum value of } \sin \theta \text{ is } 1 \text{ and maximum value of } \cos \theta \text{ is } 1.$$

$$\Rightarrow \sin \alpha = \cos \beta = 1$$

$$\Rightarrow \alpha = 90^\circ, \beta = 0^\circ$$

$$\sin \left(\frac{2\alpha + \beta}{3} \right) = \sin \left(\frac{180^\circ}{3} \right)$$

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 30^\circ = \cos \left(\frac{\alpha}{3} \right)$$

$$38. (c) \cos^4 \theta - \sin^4 \theta = \frac{2}{3}$$

$$(\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta) = \frac{2}{3}$$

$$\Rightarrow \cos^2 \theta - \sin^2 \theta = \frac{2}{3}$$

$$\Rightarrow \cos^2 \theta - (1 - \cos^2 \theta) = \frac{2}{3}$$

$$\Rightarrow 2\cos^2 \theta - 1 = \frac{2}{3}$$

$$39. (a) \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} = \frac{1 - \sin \theta + 1 + \sin \theta}{1 - \sin^2 \theta}$$

$$= \frac{2}{\cos^2 \theta} = 2 \sec^2 \theta$$

$$40. (a) \cos^2 \theta (1 + \tan^2 \theta) = \cos^2 \theta \left(1 + \frac{\sin^2 \theta}{\cos^2 \theta} \right)$$

$$= \cos^2 \theta + \sin^2 \theta = 1$$

$$41. (d) (\sec^2 \theta - \tan^2 \theta) = 1$$

trigonometric identity and hence it is valid for all values of θ .

$$42. (b) \tan x = \frac{4}{3} \sqrt{\frac{(1 - \sin x)(1 + \sin x)}{(1 + \cos x)(1 - \cos x)}}$$

$$= \sqrt{\frac{1 - \sin^2 x}{1 - \cos^2 x}} = \sqrt{\frac{\cos^2 x}{\sin^2 x}} = \frac{\cos x}{\sin x} = \cot x = \frac{3}{4}$$

$$43. (a) \sec^4 \theta - \tan^4 \theta = (\sec^2 \theta + \tan^2 \theta)(\sec^2 \theta - \tan^2 \theta)$$

$$= \sec^2 \theta + \tan^2 \theta = \frac{7}{12}$$

$$44. (b) \sin^2 60^\circ + \cos^2 (3x - 9^\circ) = 1$$

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 3x - 9^\circ = 60^\circ$$

$$3x = 69^\circ, \quad x = 23^\circ$$

$$45. (b) \sin (x - y) = \frac{1}{2} = \sin 30^\circ$$

$$x - y = 30^\circ \quad \dots (i)$$

$$\cos (x + y) = \frac{1}{2} = \cos 60^\circ$$

$$x + y = 60^\circ \quad \dots (ii)$$

Adding (i) and (ii) :

$$2x = 90^\circ$$

$$x = 45^\circ$$

... (iii)

from (ii) and (iii) : $y = 15^\circ$

$$46. (b) (\sec^2 \theta - 1)(\operatorname{cosec}^2 \theta - 1) = \tan^2 \theta \times \cot^2 \theta = 1$$

$$47. (c) 2 \cos 3\theta_1 = 1, \quad \cos 3\theta_1 = \frac{1}{2} = \cos 60^\circ$$

$$3\theta_1 = 60^\circ, \quad \theta_1 = 20^\circ, \quad 2 \sin 2\theta_2 = \sqrt{3}$$

$$\sin 2\theta_2 = \frac{\sqrt{3}}{2} = \sin 60^\circ, \quad 2\theta_2 = 60^\circ, \quad \theta_2 = 30^\circ$$

$$48. (d) \tan (\alpha + \beta) = 1; \tan (\alpha - \beta) = \frac{1}{\sqrt{3}}$$

$$\alpha + \beta = 45^\circ; \alpha - \beta = 30^\circ, \quad 2\alpha + 2\beta = 90^\circ$$

$$49. (d) -\sqrt{a^2 + b^2} \leq a \sin \theta + b \cos \theta \leq \sqrt{a^2 + b^2}$$

$$\Rightarrow \text{minimum value of } (\sin \theta + \cos \theta)$$

$$\text{is } -\sqrt{1^2 + 1^2} = -\sqrt{2}$$

$$50. (c) \sin \alpha = \sin \beta, \quad \cos \alpha = \cos \beta, \quad \sin \alpha - \sin \beta = 0$$

$$2 \cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) = 0$$

$$\cos \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) = 0 \quad \dots (i)$$

$$\cos \alpha - \cos \beta = 0$$

$$2 \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\beta - \alpha}{2} \right) = 0$$

$$- \sin \left(\frac{\alpha + \beta}{2} \right) \sin \left(\frac{\alpha - \beta}{2} \right) = 0 \quad \dots (ii)$$

from (i) and (ii)

$$\sin \left(\frac{\alpha - \beta}{2} \right) \left[\cos \left(\frac{\alpha + \beta}{2} \right) - \sin \left(\frac{\alpha + \beta}{2} \right) \right] =$$

$$\sin \left(\frac{\alpha - \beta}{2} \right) = 0 \quad \text{or} \quad \cos \left(\frac{\alpha + \beta}{2} \right) = \sin \left(\frac{\alpha + \beta}{2} \right)$$

$$\tan \left(\frac{\alpha + \beta}{2} \right) = 1, \quad \frac{\alpha + \beta}{2} = 45^\circ, \quad \alpha + \beta = 90^\circ$$

$$\sin \alpha = \sin \beta \text{ and } \alpha + \beta = 90^\circ, \quad \alpha = \beta = 45^\circ$$

$$\text{so, } \sin \left(\frac{\alpha - \beta}{\sqrt{2}} \right) = \sin (0^\circ) = 0$$

Moderate

1. (a) $\tan \theta = \frac{x}{y}, \frac{x \sin \theta + y \cos \theta}{x \sin \theta - y \cos \theta}$

$$= \frac{x \frac{\sin \theta}{\cos \theta} + y}{x \frac{\sin \theta}{\cos \theta} - y} = \frac{x \tan \theta + y}{x \tan \theta - y}$$

$$= \frac{x \times \frac{x}{y} + y}{x \times \frac{x}{y} - y} = \frac{x^2 + y^2}{x^2 - y^2}$$

2. (c) $7 \sin^2 \theta + 3 \cos^2 \theta = 4$
 $3 \sin^2 \theta + 3 \cos^2 \theta = 4 - 4 \sin^2 \theta$

$$3 \times 1 = 4 (1 - \sin^2 \theta) = \cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \frac{\sqrt{3}}{2}, \theta = 30, \tan \theta = \frac{1}{\sqrt{3}}$$

3. (d) $\sec^2 A = 3$
 $\tan^2 A = \sec^2 A - 1 = 2$

$$\cot^2 A = \frac{1}{\tan^2 A} = \frac{1}{2}$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A = 1 + \frac{1}{2} = \frac{3}{2}$$

$$\frac{\tan^2 A - (\operatorname{cosec}^2 A)}{\tan^2 A + \operatorname{cosec}^2 A} = \frac{2 - \frac{3}{2}}{2 + \frac{3}{2}} = \frac{1}{7}$$

4. (a) $\sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sqrt{\frac{1 + \sin A}{1 - \sin A} \times \frac{1 + \sin A}{1 + \sin A}}$
 $= \sqrt{\frac{(1 + \sin A)^2}{(1 - \sin^2 A)}} = \sqrt{\frac{(1 + \sin A)^2}{\cos^2 A}} = \frac{1 + \sin A}{\cos A}$
 $= \frac{1}{\cos A} + \frac{\sin A}{\cos A} = \sec A + \tan A$

5. (a) $\sec^4 A - \sec^2 A$
 $= \sec^2 A (\sec^2 A - 1) = \sec^2 A \cdot \tan^2 A$
 $= (1 + \tan^2 A) (\tan^2 A) = \tan^2 A + \tan^4 A$

6. (b) $\cot^4 \theta - \operatorname{cosec}^4 \theta + \cot^2 \theta + \operatorname{cosec}^2 \theta$
 $= \cot^4 \theta + \cot^2 \theta - \operatorname{cosec}^4 \theta + \operatorname{cosec}^2 \theta$
 $= \cot^2 \theta (\cot^2 \theta + 1) - \operatorname{cosec}^2 \theta (\operatorname{cosec}^2 \theta - 1)$
 $= \cot^2 \theta \cdot \operatorname{cosec}^2 \theta - \operatorname{cosec}^2 \theta \cdot \cot^2 \theta = 0$

7. (a) $\frac{\sin \theta}{1 - \cot \theta} + \frac{\cos \theta}{1 - \tan \theta} = \frac{\sin \theta}{1 - \frac{\cos \theta}{\sin \theta}} + \frac{\cos \theta}{1 - \frac{\sin \theta}{\cos \theta}}$

$$= \frac{\sin^2 \theta}{\sin \theta - \cos \theta} + \frac{\cos^2 \theta}{\cos \theta - \sin \theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta - \sin \theta} = \sin \theta + \cos \theta$$

8. (b) $\frac{\cot \theta}{\cot \theta - \cot 3\theta} + \frac{\tan \theta}{\tan \theta - \tan 3\theta}$

$$\frac{\frac{\cos \theta}{\sin \theta}}{\frac{\cos \theta}{\sin \theta} - \frac{\cos 3\theta}{\sin 3\theta}} + \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin \theta}{\cos \theta} - \frac{\sin 3\theta}{\cos 3\theta}}$$

$$= \frac{\frac{\cos \theta}{\sin \theta}}{\frac{\cos \theta \sin 3\theta - \sin \theta \cos 3\theta}{\sin \theta \sin 3\theta}} + \frac{\frac{\sin \theta}{\cos \theta}}{\frac{\sin \theta \cos 3\theta - \sin 3\theta \cos \theta}{\sin \theta \cos 3\theta}}$$

$$= \frac{\cos \theta}{\sin \theta} \cdot \frac{\sin \theta \sin 3\theta}{\cos \theta \sin 3\theta - \sin \theta \cos 3\theta} + \frac{\sin \theta}{\cos \theta} \cdot \frac{\cos \theta \cos 3\theta}{\sin \theta \cos 3\theta - \sin 3\theta \cos \theta}$$

$$= \frac{\cos \theta \sin 3\theta - \sin \theta \cos 3\theta}{\cos \theta \sin 3\theta - \sin \theta \cos 3\theta} = 1$$

9. (c); $\sin \theta + \cos \theta = m$
squaring on both side
 $\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cdot \cos \theta = m^2$
 $1 + 2 \sin \theta \cdot \cos \theta = m^2$

$$\sin \theta \cdot \cos \theta = \frac{m^2 - 1}{2}$$

$$\sin^3 \theta + \cos^3 \theta = n$$

$$(\sin \theta + \cos \theta)^3 - 3 \sin \theta \cdot \cos \theta (\sin \theta + \cos \theta) = n$$

Using equation (i) and (ii)

$$(m)^3 - 3 \left(\frac{m^2 - 1}{2} \right) (m) = n$$

$$2m^3 - 3(m^3 - m) = 2n, \quad 2m^3 - 3m^3 + 3m = 2n$$

$$m^3 - 3m + 2n = 0$$

10. (c); $\cot 75^\circ = \cot (90^\circ - 15^\circ) = \tan 15^\circ$
 similarly,
 $\cot 74^\circ = \tan 16^\circ$, $\cot 73^\circ = \tan 17^\circ$
 $\cot 45^\circ = 1$
 $\therefore (\cot 15^\circ \cot 75^\circ) (\cot 16^\circ \cot 74^\circ) \dots (\cot 45^\circ)$
 $= (\cot 15^\circ \cdot \tan 15^\circ) \cdot (\cot 16^\circ \cdot \tan 16^\circ) \dots (\cot 45^\circ \cdot \tan 45^\circ)$
 $= 1 \times 1 \times \dots \cot 45^\circ = 1$

11. (a); $\cos 11^\circ = \cos (90^\circ - 79^\circ) = \sin 79^\circ$
 $\sin 11^\circ = \sin (90^\circ - 79^\circ) = \cos 79^\circ$
 $\sin 79^\circ \cos 11^\circ + \sin 11^\circ \cos 79^\circ$
 $= \sin 79^\circ \cdot \sin 79^\circ + \cos 79^\circ \cdot \cos 79^\circ$
 $= \sin^2 79^\circ + \cos^2 79^\circ = 1$

12. (c) $\cos 160^\circ = \cos (180^\circ - 20^\circ) = -\cos 20^\circ$
 $\cos 140^\circ = \cos (180^\circ - 40^\circ) = -\cos 40^\circ$
 similarly, $\cos 120^\circ = -\cos 60^\circ$
 $\cos 20^\circ + \cos 40^\circ + \cos 60^\circ + \dots + \cos 160^\circ + \cos 180^\circ$
 $= (\cos 20^\circ + \cos 160^\circ) + (\cos 40^\circ + \cos 140^\circ) + (\cos 60^\circ + \cos 120^\circ) + (\cos 80^\circ + \cos 100^\circ) + \cos 180^\circ$
 $= (\cos 20^\circ - \cos 20^\circ) + (\cos 40^\circ - \cos 40^\circ) + (\cos 60^\circ - \cos 60^\circ) + (\cos 80^\circ - \cos 80^\circ) + (\cos 180^\circ)$
 $= -1$

13. (a) $\tan \frac{x}{2} = \sqrt{\frac{1 - \cos x}{1 + \cos x}}$
 for $x = 45^\circ$
 $\tan 22^\circ 30' = \sqrt{\frac{1 - \cos 45^\circ}{1 + \cos 45^\circ}}$

$$= \sqrt{\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}} = \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}} = \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1} \times \frac{\sqrt{2} - 1}{\sqrt{2} - 1}}$$

$$= \sqrt{\frac{(\sqrt{2} - 1)^2}{1}} = \sqrt{2} - 1$$

14. (a) $4 \sin 45^\circ \cos 15^\circ$
 $= 2[\sin (45^\circ + 15^\circ) + \sin (45^\circ - 15^\circ)]$
 $= 2[\sin 60^\circ + \sin 30^\circ] = 2 \left(\frac{\sqrt{3}}{2} + \frac{1}{2} \right) = \sqrt{3} + 1$

15. (b) $\sin \theta = \frac{1}{2} \Rightarrow \cos \theta = \frac{\sqrt{3}}{2}$
 $\cos \phi = 1 \Rightarrow \sin \phi = 0$
 $\cot (\theta + \phi) = \frac{\cos(\theta + \phi)}{\sin(\theta + \phi)} = \frac{\cos \theta \cos \phi - \sin \theta \sin \phi}{\sin \theta \cos \phi + \cos \theta \sin \phi}$
 $= \frac{\frac{\sqrt{3}}{2} \times 1 - \frac{1}{2} \times 0}{\frac{1}{2} \times 1 + \frac{\sqrt{3}}{2} \times 0} = \frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} = \sqrt{3}$

16. (d) ABCD is cyclic quadrilateral
 $A + C = B + D = \pi$
 $A = \pi - C \Rightarrow \cos A = \cos (\pi - C) = -\cos C$
 $\cos A + \cos C = 0$
 similarly, $\cos B + \cos D = 0$
 $\cos A + \cos C - (\cos B + \cos D) = 0$
 $= \cos A - \cos B + \cos C - \cos D = 0$

17. (c) $\frac{1 + \tan 15^\circ}{1 - \tan 15^\circ} = \frac{\tan 45^\circ + \tan 15^\circ}{1 - \tan 45^\circ \cdot \tan 15^\circ}$
 $= \tan (45^\circ + 15^\circ) = \tan 60^\circ = \sqrt{3}$

$$\left[\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} \right]$$

18. (d) $\cos A - \sin A > 0$, $\sin A < \cos A$, $\tan A < 1$, $A < 45^\circ$
 $\therefore \sin A + \cos A < \sin 45^\circ + \cos 45^\circ = \sqrt{2}$

19. (a) $\frac{\cot A + (\operatorname{cosec} A - 1)}{\cot A - (\operatorname{cosec} A - 1)} \times \frac{\cot A + (\operatorname{cosec} A - 1)}{\cot A + (\operatorname{cosec} A - 1)}$
 $= \frac{\cot^2 A + (\operatorname{cosec} A - 1)^2 + 2 \cot A (\operatorname{cosec} A - 1)}{\cot^2 A - (\operatorname{cosec} A - 1)^2}$
 $= \frac{(\operatorname{cosec}^2 A - 1) + (\operatorname{cosec} A - 1)^2 + 2 \cot A (\operatorname{cosec} A - 1)}{\cot^2 A - [\operatorname{cosec}^2 A + 1 - 2 \operatorname{cosec} A]}$
 $= \frac{2(\operatorname{cosec} A - 1)(\operatorname{cosec} A + \cot A)}{2(\operatorname{cosec} A - 1)}$
 $= \operatorname{cosec} A + \cot A$

20. (a); $\cos 3\theta + \sin 3\theta = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 3\theta + \frac{1}{\sqrt{2}} \sin 3\theta \right)$
 $= \sqrt{2} (\sin 45^\circ \cos 3\theta + \cos 45^\circ \sin 3\theta)$
 $= \sqrt{2} \sin (45^\circ + 3\theta)$
 maximum value occurs when,
 $\sin (45^\circ + 3\theta) = 1$
 $\Rightarrow 3\theta = 45^\circ$ or $\theta = 15^\circ$

21. (a); Given $= \frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ - \sin 11^\circ}$

$$\frac{\cos 11^\circ + \sin 11^\circ}{\cos 11^\circ} = \frac{1 + \tan 11^\circ}{1 - \tan 11^\circ}$$

$$\cong \frac{\cos 11^\circ - \sin 11^\circ}{\cos 11^\circ}$$

$$\tan (A + B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B} \quad (\because \tan 45^\circ = 1)$$

$$\cong \tan (45 + 11) = \tan 56^\circ$$

$$\begin{aligned}
 22. \quad (d) \quad & \sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - \sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right) \\
 &= \frac{1}{2} \left[2\sin^2\left(\frac{\pi}{8} + \frac{A}{2}\right) - 2\sin^2\left(\frac{\pi}{8} - \frac{A}{2}\right) \right] \\
 &= \frac{1}{2} \left[1 - \cos\left(\frac{\pi}{4} + A\right) - 1 + \cos\left(\frac{\pi}{4} - A\right) \right]
 \end{aligned}$$

$$\begin{aligned}
 & [\because 2\sin^2\theta = 1 - \cos 2\theta] \\
 &= \frac{1}{2} \left[2\sin\frac{\pi}{4}\sin A \right] = \frac{1}{\sqrt{2}}\sin A
 \end{aligned}$$

23. (a) Clearly, $\sin\theta$ is increasing in 1st quadrant and

$$0 < \sin\theta < \cos\theta \text{ when } 0 < \theta < \frac{\pi}{4}.$$

So, statements 1. and 3. are correct.

clearly, statement 4 is incorrect,
Since,

$$\sin\frac{\pi}{3} = \frac{\sqrt{3}}{2} > \frac{1}{2} = \cos\frac{\pi}{3}$$

$$24. \quad (c) \quad \tan x = \frac{1}{\sqrt{3}}, x = 30^\circ$$

$$\sin y = \frac{1}{\sqrt{2}}, y = 45^\circ$$

$$x + y = 30^\circ + 45^\circ = 75^\circ$$

So, $x + y$ lies between 0° and 90° .

$$\begin{aligned}
 25. \quad (d) \quad & \cot 70^\circ = \cot(90^\circ - 20^\circ) = \tan 20^\circ \\
 & \cot 80^\circ = \cot(90^\circ - 10^\circ) = \tan 10^\circ \\
 & \text{so,} \\
 & \cot 10^\circ \cdot \cot 80^\circ \cdot \cot 20^\circ \cdot \cot 70^\circ \cdot \cot 60^\circ \\
 &= (\cot 10^\circ \cdot \tan 10^\circ) \cdot (\cot 20^\circ \cdot \tan 20^\circ) \cdot \cot 60^\circ \\
 &= 1 \cdot 1 \cdot \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}
 \end{aligned}$$

$$\begin{aligned}
 26. \quad (a) \quad & \sin \alpha \sec(30^\circ + \alpha) = 1, \quad \frac{\sin \alpha}{\cos(30^\circ + \alpha)} = 1 \\
 & \sin \alpha = \cos(30^\circ + \alpha) = \sin(90^\circ - 30^\circ + \alpha) \\
 & \sin \alpha = \sin(60^\circ - \alpha) \\
 \Rightarrow & \alpha = 60^\circ - \alpha \quad \text{or,} \quad \alpha = 30^\circ \\
 & \sin \alpha + \cos 2\alpha = \sin 30^\circ + \cos 60^\circ = \frac{1}{2} + \frac{1}{2} = 1
 \end{aligned}$$

$$\begin{aligned}
 27. \quad (b) \quad & \cos^2\theta + \cos^4\theta = 1, \quad 1 - \cos^2\theta = \cos^4\theta \\
 & \sin^2\theta = \cos^4\theta, \quad \tan^2\theta = \cos^2\theta \\
 & \tan^2\theta + \tan^4\theta = \cos^2\theta + \cos^4\theta = 1
 \end{aligned}$$

$$\begin{aligned}
 28. \quad (b) \quad & \tan 86^\circ = \tan(90^\circ - 4^\circ) = \cot 4^\circ \\
 & \tan 43^\circ = \cot 47^\circ, \quad \tan 4^\circ \cdot \tan 43^\circ \cdot \tan 47^\circ \cdot \tan 86^\circ \\
 &= (\tan 4^\circ \cdot \tan 86^\circ) (\tan 43^\circ \cdot \tan 47^\circ) \\
 &= (\tan 4^\circ \cdot \cot 4^\circ) \cdot (\tan 47^\circ \cdot \cot 47^\circ) = 1 \times 1 = 1
 \end{aligned}$$

$$29. \quad (d) \quad \sin 17^\circ = \frac{x}{y}$$

$$\cos 17^\circ = \frac{\sqrt{1 - \frac{x^2}{y^2}}}{\frac{y}{y}} = \frac{\sqrt{y^2 - x^2}}{y}$$

$$\sec 17^\circ = \frac{y}{\sqrt{y^2 - x^2}}$$

$$\sin 73^\circ = \sin(90^\circ - 17^\circ) = \cos 17^\circ$$

$$\sin 73^\circ = \frac{\sqrt{y^2 - x^2}}{y}$$

$$\text{so, } \sec 17^\circ = \sin 73^\circ$$

$$\begin{aligned}
 &= \frac{y}{\sqrt{y^2 - x^2}} = \frac{\sqrt{y^2 - x^2}}{y} = \frac{y^2 - (y^2 - x^2)}{y\sqrt{y^2 - x^2}} \\
 &= \frac{x^2}{y\sqrt{y^2 - x^2}}
 \end{aligned}$$

$$30. \quad (b); \quad \frac{1}{\cos x} = \frac{1}{\sin y}$$

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - \cos^2 x} = \sqrt{\sin^2 x} = \sin x.$$

$$\sin(x + y) = \sin x \cdot \cos y + \cos x \cdot \sin y.$$

$$= \sin^2 x + \cos^2 x = 1.$$

$$31. \quad (a); \quad \tan 89^\circ = \tan(90^\circ - 1^\circ) = \cot 1^\circ$$

$$\text{similarly, } \tan 88^\circ = \cot 2^\circ;$$

$$\tan 87^\circ = \cot 3^\circ, \dots\dots$$

$$\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots\dots \tan 89^\circ =$$

$$(\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \dots\dots \tan 45^\circ \dots\dots \tan 87^\circ \cdot$$

$$\tan 88^\circ \cdot \tan 89^\circ.$$

$$= (\tan 1^\circ \cdot \tan 89^\circ) \cdot (\tan 2^\circ \cdot \tan 88^\circ) \cdot (\tan 3^\circ \cdot \tan 87^\circ) \dots\dots \tan 45^\circ.$$

$$= (\tan 1^\circ \cdot \cot 1^\circ) \cdot (\tan 2^\circ \cdot \cot 2^\circ) \cdot (\tan 3^\circ \cdot \cot 3^\circ)$$

$$= 1$$

$$32. \quad (c) \quad A + B + C = \pi$$

$$\frac{A+B}{2} = \frac{\pi-C}{2} = \frac{\pi}{2} - \frac{C}{2}$$

$$\sin\left(\frac{A+B}{2}\right) = \sin\left(\frac{\pi}{2} - \frac{C}{2}\right)$$

$$= \sin\left(\frac{A+B}{2}\right) = \cos\frac{C}{2}$$

$$\cos\left(\frac{A+B}{2}\right) = \cos\left(\frac{\pi}{2} - \frac{C}{2}\right) = \sin\frac{C}{2}$$

$$\tan\left(\frac{A+B}{2}\right) = \tan\left(\frac{\pi}{2} - \frac{C}{2}\right) = \cot\frac{C}{2} \neq \sec\frac{C}{2}$$

$$\cot\left(\frac{A+B}{2}\right) = \cot\left(\frac{\pi}{2} - \frac{C}{2}\right) = \tan\frac{C}{2}$$

$$\begin{aligned}
 33. \quad (c) \quad & (\sec A - \cos A)^2 + (\operatorname{cosec} A - \sin A)^2 - (\cot A - \tan A)^2 \\
 &= \sec^2 A + \cos^2 A - 2\sec A \cdot \cos A + \operatorname{cosec}^2 A + \sin^2 A - 2\operatorname{cosec} A \cdot \sin A - [\cot^2 A + \tan^2 A - 2\cot A \cdot \tan A] \\
 &= (\sec^2 A - \tan^2 A) + (\cos^2 A + \sin^2 A) + (\operatorname{cosec}^2 A - \cot^2 A) - 2 - 2 + 2 \\
 &= 1 + 1 + 1 - 2 = 1
 \end{aligned}$$

$$\begin{aligned}
 34. \quad (d) \quad & 7 \sin^2 \theta + 3 \cos^2 \theta = 4 \\
 & 4 \sin^2 \theta + 3 \sin^2 \theta + 3 \cos^2 \theta = 4 \\
 & 4 \sin^2 \theta + 3 = 4, \quad 4 \sin^2 \theta = 1 \\
 & \sin \theta = \frac{1}{2} \quad [\because \theta \text{ in an acute angle}] \\
 & \Rightarrow \operatorname{cosec} \theta = \frac{1}{\sin \theta} = 2
 \end{aligned}$$

$$\begin{aligned}
 35. \quad (a) \quad & \cot 18^\circ \cot 72^\circ \cos^2 22^\circ + \frac{\cot 18^\circ}{\tan 72^\circ \sec^2 68^\circ} \\
 &= \cot 18^\circ \cdot \tan 18^\circ \cdot \cos^2 22^\circ + \left(\frac{\cot 18^\circ}{\cot 18^\circ} \right) \times \cos^2 68^\circ \\
 &= \cos^2 22^\circ + \cos^2 68^\circ, \quad = \cos^2 22^\circ + \sin^2 22^\circ = 1
 \end{aligned}$$

$$\begin{aligned}
 36. \quad (c) \quad & \frac{1}{\operatorname{cosec}^2 51^\circ} + \sin^2 39^\circ + \tan^2 51^\circ \\
 &= \frac{1}{\sin^2 51^\circ \cdot \sec^2 39^\circ} \\
 &= \sin^2 51^\circ + \sin^2 39^\circ + \tan^2 (90^\circ - 39^\circ) - \frac{1}{\sin^2 (90^\circ - 39^\circ) \cdot \sec^2 39^\circ} \\
 &= \cos^2 39^\circ + \sin^2 39^\circ + \cot^2 39^\circ - \frac{1}{\cos^2 39^\circ \cdot \sec^2 39^\circ} \\
 &= 1 + \cot^2 39^\circ - 1, \quad = \operatorname{cosec}^2 39^\circ - 1 = x^2 - 1
 \end{aligned}$$

$$\begin{aligned}
 37. \quad (b) \quad & \text{By componendo and dividendo :} \\
 & \frac{\tan \theta + \cot \theta}{\tan \theta - \cot \theta} = \frac{\tan \theta + \cot \theta + \tan \theta - \cot \theta}{\tan \theta + \cot \theta - \tan \theta + \cot \theta} = \frac{2 + 1}{2 - 1} \\
 & \frac{2 \tan \theta}{2 \cot \theta} = 3, \quad \frac{\sin \theta}{\cos \theta} \times \frac{\sin \theta}{\cos \theta} = 3 \\
 & \text{or, } \sin^2 \theta = 3 \cos^2 \theta, \quad \sin^2 \theta = 3(1 - \sin^2 \theta) \\
 & 4 \sin^2 \theta = 3 \\
 & \sin \theta = \frac{\sqrt{3}}{2} \quad [\text{as } 0^\circ \leq \theta \leq 90^\circ]
 \end{aligned}$$

$$\begin{aligned}
 38. \quad (d) \quad & \sin \theta + \operatorname{cosec} \theta = 2, \quad \sin \theta + \frac{1}{\sin \theta} = 2 \\
 & \sin^2 \theta - 2 \sin \theta + 1 = 0, \quad (\sin \theta - 1)^2 = 0, \sin \theta = 1 \\
 & \text{so, } \sin^5 \theta + \operatorname{cosec}^5 \theta = \sin^5 \theta + \frac{1}{\sin^5 \theta}, = 1 + 1 = 2
 \end{aligned}$$

$$\begin{aligned}
 39. \quad (b); \quad & \cos^2 5^\circ + \cos^2 10^\circ + \cos^2 15^\circ + \dots + \cos^2 80^\circ + \cos^2 85^\circ + \cos^2 90^\circ \\
 &= (\cos^2 5^\circ + \cos^2 85^\circ) + (\cos^2 10^\circ + \cos^2 80^\circ) + \dots + \cos^2 40^\circ + \cos^2 50^\circ + \cos^2 45^\circ + \cos^2 90^\circ \\
 &= 1 + 1 + \dots \text{ 8 times } + \frac{1}{2} + 0 = 8 \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 40. \quad (c); \quad & 2 \cos \theta - \sin \theta = \frac{1}{\sqrt{2}} \\
 & \text{Let } 2 \sin \theta + \cos \theta = x \\
 & \text{on squaring and adding :} \\
 & 4 \cos^2 \theta + \sin^2 \theta - 4 \sin \theta \cos \theta + 4 \sin^2 \theta + \cos^2 \theta + 4 \sin \theta \cdot \cos \theta = \frac{1}{2} + x^2
 \end{aligned}$$

$$4(\cos^2 \theta + \sin^2 \theta) + (\cos^2 \theta + \sin^2 \theta) = \frac{1}{2} + x^2$$

$$4 + 1 = \frac{1}{2} + x^2, \quad x^2 = 5 - \frac{1}{2}, \quad x^2 = \frac{9}{2} = \frac{3}{\sqrt{2}}$$

$$\begin{aligned}
 41. \quad (c); \quad & \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 3 \\
 & 4 \cos \theta = 2 \sin \theta, \quad \tan \theta = 2, \quad \sin^4 \theta - \cos^4 \theta, \\
 & (\sin^2 \theta + \cos^2 \theta)(\sin^2 \theta - \cos^2 \theta) \\
 & \Rightarrow \cos^2 \theta (\tan^2 \theta - 1) \\
 & \Rightarrow \frac{3}{\sec^2 \theta} \Rightarrow \frac{3}{1 + \tan^2 \theta} = \frac{3}{1 + 4} = \frac{3}{5}
 \end{aligned}$$

$$\begin{aligned}
 42. \quad (d) \quad & \frac{\sin 39^\circ}{\cos 51^\circ} + 2 \tan 11^\circ \cdot \tan 79^\circ \\
 & \tan 31^\circ \cdot \tan 59^\circ \tan 45^\circ - 3(\sin^2 21^\circ + \sin^2 69^\circ) \\
 &= \frac{\sin 39^\circ}{\cos(90^\circ - 39^\circ)} + 2 \tan 11^\circ \times \tan(90^\circ - 11^\circ) \cdot \tan 31^\circ \cdot \tan(90^\circ - 31^\circ) \cdot 1 - 3[\sin^2 21^\circ + \sin^2(90^\circ - 21^\circ)] \\
 &= \frac{\sin 39^\circ}{\sin 39^\circ} + 2 \tan 11^\circ \cdot \cot 11^\circ \cdot \tan 31^\circ \cdot \cot 31^\circ - 3(\sin^2 21^\circ + \cos^2 21^\circ) = 1 + 2 - 3 = 0
 \end{aligned}$$

$$\begin{aligned}
 43. \quad (c) \quad & A = \tan 11^\circ \times \tan 29^\circ, \\
 & B = 2 \cot 61^\circ \cot 79^\circ \\
 &= 2 \cot(90^\circ - 29^\circ) \cot(90^\circ - 11^\circ) \\
 &= 2 \tan 29^\circ \times \tan 11^\circ = 2A
 \end{aligned}$$

$$\begin{aligned}
 44. \quad (b) \quad & \cos^2 \alpha + \cos^2 \beta = 2 \\
 & 1 - \sin^2 \alpha + 1 - \sin^2 \beta = 2 \\
 & \sin^2 \alpha + \sin^2 \beta = 0, \quad \sin \alpha = \sin \beta = 0 \\
 & \alpha = \beta = 0, \quad \tan^3 \alpha + \sin^5 \beta = 0
 \end{aligned}$$

$$\begin{aligned}
 45. \quad (a) \quad & \sin \theta - \cos \theta = \frac{7}{13} \quad \dots(i) \\
 & \text{Let } \sin \theta + \cos \theta = x \quad \dots(ii) \\
 & \text{on squaring both equations and adding :}
 \end{aligned}$$

- $2(\sin^2\theta + \cos^2\theta) = \frac{49}{169} + x^2$
 $x^2 = 2 - \frac{49}{169} = \frac{338-49}{169}, \quad x^2 = \frac{289}{169}, \quad x = \frac{17}{13}$
46. (d) $\sin(2x - 20^\circ) = \cos(2y + 20^\circ)$
 $\sin(2x - 20^\circ) = \sin(90^\circ - 2y - 20^\circ)$
 $2x - 20^\circ = 70^\circ - 2y$
 $2x + 2y = 90^\circ$
 $x + y = 45^\circ$
 $\tan(x + y) = \tan 45^\circ = 1.$
47. (d) $(\sec x \cdot \sec y + \tan x \cdot \tan y)^2 - (\sec x \tan y + \tan x \cdot \sec y)^2$
 $= \sec^2 x \cdot \sec^2 y + \tan^2 x \cdot \tan^2 y + 2 \sec x \cdot \sec y \cdot \tan x \cdot \tan y - \sec^2 x \tan^2 y - \tan^2 x \sec^2 y - 2 \sec x \tan y \cdot \tan x \cdot \sec y$
 $= \sec^2 x \cdot \sec^2 y + \tan^2 x \cdot \tan^2 y - \sec^2 x \tan^2 y - \tan^2 x \sec^2 y$
 $= \sec^2 x (\sec^2 y - \tan^2 y) - \tan^2 x (\sec^2 y - \tan^2 y)$
 $= \sec^2 x - \tan^2 x = 1$

48. (b) x^{th} term,
 $a_n = a + (n-1)d, \quad 85 = 5 + (n-1)1$
 $80 = n-1 \quad n = 81$
 $\Rightarrow 80 \text{ terms} + 1 \text{ middle term.}$
 $\text{So, } \sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 45^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ$
 $= (\sin^2 5^\circ + \sin^2 85^\circ) + \dots + (\sin^2 6^\circ + \sin^2 84^\circ) + \dots$
 $\text{upto } \dots \text{ to } 40 \text{ terms} + \sin^2 45^\circ$
 $= (\sin^2 5^\circ + \cos^2 5^\circ) + (\sin^2 6^\circ + \cos^2 6^\circ) + \dots \text{ to } 40 \text{ terms} + \sin^2 45^\circ$
 $= 40 + \frac{1}{2} = 40\frac{1}{2}$
49. (b) $\sin \theta + \operatorname{cosec} \theta = 2, \quad \sin \theta + \frac{1}{\sin \theta} = 2$
 $(\sin \theta - 1)^2 = 0, \quad \sin \theta - 1 = 0$
 $\sin \theta = 1 \Rightarrow \operatorname{cosec} \theta \Rightarrow = 1$
 $\text{So, } \sin^{100} \theta + \operatorname{cosec}^{100} \theta, = 1 + 1 = 2$
50. (b) $\tan 7^\circ \cdot \tan 23^\circ \tan 60^\circ \cdot \tan 67^\circ \cdot \tan 83^\circ$
 $= \tan 7^\circ \cdot \tan 83^\circ \cdot \tan 23^\circ \cdot \tan 67^\circ \cdot \tan 60^\circ$
 $= (\tan 7^\circ \cdot \cot 7^\circ) \cdot (\tan 23^\circ \cdot \cot 23^\circ) \cdot \tan 60^\circ$
 $= 1 \cdot 1 \cdot \sqrt{3} = \sqrt{3}.$

Difficult

1. (a) $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$
 $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
 $\frac{1 + \cos \theta}{\sin \theta} = \frac{2 \cos^2 \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \cot \frac{\theta}{2}$
 $\text{Or, } \cot \frac{\theta}{2} = \frac{1 + \cos \theta}{\sin \theta}$
 $\text{Putting } \theta = 165^\circ$
 $\cot \left(\frac{165^\circ}{2} \right) = \frac{1 + \cos 165^\circ}{\sin 165^\circ}$
 $\cot 82\frac{1}{2}^\circ = \frac{1 - \cos 15^\circ}{\sin 15^\circ} = \frac{1 - \frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}}$
 $= \frac{2\sqrt{2} - \sqrt{3} - 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$
 $= \frac{2\sqrt{6} + 2\sqrt{2} - 3 - \sqrt{3} - \sqrt{3} - 1}{3 - 1}$
 $= \frac{2\sqrt{6} + 2\sqrt{2} - 4 - 2\sqrt{3}}{2} = \sqrt{6} + \sqrt{2} - 2 - \sqrt{3}$
2. (a) $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}; \quad \sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
 $\therefore \tan \frac{\theta}{2} = \frac{1 - \cos \theta}{\sin \theta}$
 $\text{Putting } \theta = 285^\circ$
 $\tan \frac{285^\circ}{2} = \frac{1 - \cos 285^\circ}{\sin 285^\circ} = \frac{1 - \sin 15^\circ}{-\cos 15^\circ}$
 $= \frac{1 - (\sqrt{3}-1)}{2\sqrt{2}} = -\frac{(2\sqrt{2} - \sqrt{3} + 1)}{\sqrt{3} + 1}$
 $= -\left(\frac{2\sqrt{2} - \sqrt{3} + 1}{\sqrt{3} + 1} \times \frac{\sqrt{3} - 1}{\sqrt{3} - 1} \right)$
 $= -\left(\frac{2\sqrt{6} - 2\sqrt{2} - 3 + \sqrt{3} + \sqrt{3} - 1}{3 - 1} \right)$
 $= -\left(\frac{2\sqrt{6} - 2\sqrt{2} - 4 + 2\sqrt{3}}{2} \right)$
 $= -(\sqrt{6} - \sqrt{2} - 2 + \sqrt{3}) = (\sqrt{2} + 2 - \sqrt{3} - \sqrt{6})$
3. (c) $\text{LHS} = \frac{1}{2 \sin \alpha} (2 \sin \alpha \cos \alpha) \cos 2\alpha$
 $\cos 3\alpha \cdot \cos 4\alpha \cdot \cos 5\alpha \cdot \cos 6\alpha$

$$\begin{aligned}
&= \frac{2 \sin 2\alpha \cdot \cos 2\alpha}{2 \times 2 \sin \alpha} \cos 3\alpha \cdot \cos 4\alpha \cdot \cos 5\alpha \cdot \cos 6\alpha \\
&= \frac{2 \sin 4\alpha \cos 4\alpha}{2 \times 4 \sin \alpha} \cos 3\alpha \cdot \cos 5\alpha \cdot \cos 6\alpha \\
&= \frac{\sin 8\alpha \cos 3\alpha \cos 5\alpha \cos 6\alpha}{8 \sin \alpha}
\end{aligned}$$

$$\text{But, } \alpha = \frac{\pi}{13} \Rightarrow 13\alpha = \pi \Rightarrow 8\alpha = \pi - 5\alpha$$

$$\begin{aligned}
\therefore & \frac{\sin(\pi - 5\alpha) \cos 3\alpha \cos 5\alpha \cos 6\alpha}{8 \sin \alpha} \\
&= \frac{(2 \sin 5\alpha \cos 5\alpha) \cos 3\alpha \cos 6\alpha}{2 \times 8 \sin \alpha} \\
&= \frac{\sin 10\alpha \cos 3\alpha \cos 6\alpha}{16 \sin \alpha} \\
&= \frac{\sin(\pi - 3\alpha) \cdot \cos 3\alpha \cdot \cos 6\alpha}{16 \sin \alpha} \\
&= \frac{(2 \sin 3\alpha \cdot \cos 3\alpha) \cos 6\alpha}{2 \times 16 \sin \alpha} \\
&= \frac{2 \times \sin 6\alpha \cdot \cos 6\alpha}{2 \times 32 \sin \alpha} \\
&= \frac{\sin 12\alpha}{64 \sin \alpha} = \frac{\sin(\pi - \alpha)}{64 \sin \alpha} = \frac{\sin \alpha}{64 \sin \alpha} = \frac{1}{64}
\end{aligned}$$

$$\begin{aligned}
4. \quad (a); & \sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \cdot \sin \frac{7\pi}{14} \cdot \\
& \sin \frac{9\pi}{14} \cdot \sin \frac{11\pi}{14} \cdot \sin \frac{13\pi}{14} \\
&= \sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \cdot \sin \frac{\pi}{2} \\
& \sin \left(\pi - \frac{5\pi}{14} \right) \cdot \sin \left(\pi - \frac{3\pi}{14} \right) \cdot \sin \left(\pi - \frac{\pi}{14} \right) \\
&= \sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \cdot 1 \cdot \sin \frac{5\pi}{14} \\
& \quad \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{\pi}{14} \\
&= \left(\sin \frac{\pi}{14} \cdot \sin \frac{3\pi}{14} \cdot \sin \frac{5\pi}{14} \right)^2 \\
&= \left\{ \cos \left(\frac{\pi}{2} - \frac{\pi}{14} \right) \cdot \cos \left(\frac{\pi}{2} - \frac{3\pi}{14} \right) \cdot \cos \left(\frac{\pi}{2} - \frac{5\pi}{14} \right) \right\}^2 \\
&= \left(\cos \frac{7\pi - \pi}{14} \cdot \cos \frac{7\pi - 3\pi}{14} \cdot \cos \frac{7\pi - 5\pi}{14} \right)^2
\end{aligned}$$

$$\begin{aligned}
&= \left(\cos \frac{3\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{\pi}{7} \right)^2 \\
&= \left[\frac{2 \sin \frac{\pi}{7}}{2 \sin \frac{\pi}{7}} \cdot \cos \frac{\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{3\pi}{7} \right]^2 \\
&= \left(\frac{2}{2} \cdot \frac{\sin \frac{2\pi}{7} \cdot \cos \frac{2\pi}{7} \cdot \cos \frac{3\pi}{7}}{2 \sin \frac{\pi}{7}} \right)^2 \\
&= \left(\frac{\sin \frac{4\pi}{7} \cdot \cos \left(\pi - \frac{4\pi}{7} \right)}{4 \sin \frac{\pi}{7}} \right)^2 \\
&= \left(\frac{-\sin \frac{4\pi}{7} \cdot \cos \frac{4\pi}{7}}{4 \sin \frac{\pi}{7}} \right)^2 \\
&= \left(\frac{-2 \sin \frac{4\pi}{7} \cdot \cos \frac{4\pi}{7}}{2 \cdot 4 \sin \frac{\pi}{7}} \right)^2 = \left(\frac{-\sin \frac{8\pi}{7}}{8 \sin \frac{\pi}{7}} \right)^2 \\
&= \left(\frac{-\sin \left(\pi + \frac{\pi}{7} \right)}{8 \sin \frac{\pi}{7}} \right)^2 = \left(\frac{\sin \frac{\pi}{7}}{8 \sin \frac{\pi}{7}} \right)^2 = \frac{1}{64}
\end{aligned}$$

$$\begin{aligned}
5. \quad (c); & \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \left(\pi - \frac{3\pi}{8} \right) + \sin^4 \left(\pi - \frac{\pi}{8} \right) \\
&= \sin^4 \frac{\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{3\pi}{8} + \sin^4 \frac{\pi}{8} \\
&= 2 \left\{ \left(\frac{1 - \cos \frac{\pi}{4}}{2} \right)^2 + \left(\frac{1 - \cos \frac{3\pi}{4}}{2} \right)^2 \right\} \\
& \quad \left(\because \sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2} \right) \\
&= \frac{2}{2^2} \left\{ \left(1 - \frac{1}{\sqrt{2}} \right)^2 + \left(1 + \frac{1}{\sqrt{2}} \right)^2 \right\} \\
&= \frac{1}{2} \left\{ 2 \left(1 + \frac{1}{2} \right) \right\} = 1 + \frac{1}{2} = \frac{3}{2}
\end{aligned}$$

6. (d) $\sec \theta = x + \frac{1}{4x}$

$$\frac{h}{b} = \frac{4x^2 + 1}{4x}$$

$$\therefore p = \sqrt{(4x^2 + 1)^2 - (4x)^2}$$

$$= \sqrt{16x^4 + 1 + 8x^2 - 16x^2}$$

$$= \sqrt{16x^4 - 8x^2 + 1^2}$$

$$= \sqrt{(4x^2 - 1)^2}$$

$$\therefore \sec \theta + \tan \theta$$

$$= \frac{4x^2 + 1}{4x} + \frac{4x^2 - 1}{4x}$$

$$= \frac{8x^2}{4x} = 2x$$

7. (c) $(a^2 - b^2) \sin \theta + 2ab \cos \theta = a^2 + b^2$
Dividing both sides by $a^2 + b^2$,

$$\frac{(a^2 - b^2)}{(a^2 + b^2)} \sin \theta + \frac{2ab}{(a^2 + b^2)} \cos \theta = \frac{a^2 + b^2}{a^2 + b^2}$$

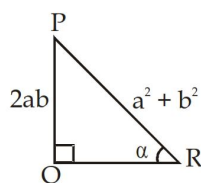
$$\Rightarrow \frac{(a^2 - b^2)}{(a^2 + b^2)} \sin \theta + \frac{2ab}{(a^2 + b^2)} \cos \theta = 1 \quad \dots(i)$$

Let $\angle R = \alpha$

$$\therefore (PR)^2 = (PQ)^2 + (QR)^2$$

$$(a^2 + b^2)^2 = (2ab)^2 + (a^2 - b^2)^2$$

both relation are right



$$\therefore \cos \alpha = \frac{a^2 - b^2}{a^2 + b^2}, \sin \alpha = \frac{2ab}{a^2 + b^2} \text{ [from equation(i)]}$$

$$\cos \alpha \cdot \sin \alpha + \sin \alpha \cdot \cos \alpha = 1$$

$$\sin(\theta + \alpha) = \sin 90^\circ$$

$$\therefore \theta + \alpha = 90^\circ$$

$$\therefore \tan \theta = \tan(90^\circ - \alpha) = \cot \alpha$$

$$= \frac{a^2 - b^2}{2ab} = \frac{1}{2ab}(a^2 - b^2)$$

8. (a) $2 - \cos^2 \theta = 3 \sin \theta \cdot \cos \theta$
dividing both sides by $\cos^2 \theta$,

$$\frac{2 - \cos^2 \theta}{\cos^2 \theta} = \frac{3 \sin \theta \cdot \cos \theta}{\cos^2 \theta}$$

$$2 \sec^2 \theta - 1 = 3 \tan \theta$$

$$2(1 + \tan^2 \theta) - 1 = 3 \tan \theta$$

$$2 + 2 \tan^2 \theta - 1 = 3 \tan \theta$$

$$1 + 2 \tan^2 \theta = 3 \tan \theta$$

$$2 \tan^2 \theta - 3 \tan \theta + 1 = 0$$

$$2 \tan^2 \theta - 2 \tan \theta - \tan \theta + 1 = 0$$

$$2 \tan \theta (\tan \theta - 1) - 1 (\tan \theta - 1) = 0$$

$$(2 \tan \theta - 1)(\tan \theta - 1) = 0$$

$$\therefore \tan \theta = 1 \text{ or } \frac{1}{2}$$

9. (c); $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$

$$\text{So, } \sin^3 \theta = \frac{3 \sin \theta - \sin 3\theta}{4}$$

LHS

$$= \frac{3 \sin \theta - \sin 3\theta}{4} + \frac{3 \sin(120^\circ + \theta) - \sin 3(120^\circ + \theta)}{4} +$$

$$+ \frac{3 \sin(240^\circ + \theta) - \sin 3(240^\circ + \theta)}{4}$$

$$= \frac{3}{4} \{ \sin \theta + \sin(120^\circ + \theta) + \sin(240^\circ + \theta) \}$$

$$- \frac{1}{4} \{ \sin 3\theta + \sin(360^\circ + 3\theta) + \sin(720^\circ + 3\theta) \}$$

$$= \frac{3}{4} \left\{ \sin \theta + 2 \sin \left(\frac{120^\circ + \theta + 240^\circ + \theta}{2} \right) \right\}$$

$$\cos \left(\frac{120^\circ + \theta - 240^\circ - \theta}{2} \right) \left\{ - \frac{1}{4} \{ \sin 3\theta + \sin 3\theta + \sin 3\theta \} \right\}$$

$$= \frac{3}{4} \{ \sin \theta + 2 \sin(180^\circ + \theta) \cos 60^\circ \} - \frac{1}{4} \{ 3 \sin 3\theta \}$$

$$= \frac{3}{4} \left\{ \sin \theta - 2 \cdot \sin \theta \cdot \frac{1}{2} \right\} - \frac{3}{4} \sin 3\theta = - \frac{3}{4} \sin 3\theta$$

10. (d) $\cot \theta + \cot(60^\circ + \theta) + \cot(120^\circ + \theta)$

$$= \cot \theta + \frac{\cot 60^\circ \cot \theta - 1}{\cot \theta + \cot 60^\circ} + \frac{\cot 120^\circ \cot \theta - 1}{\cot \theta + \cot 120^\circ}$$

$$= \cot \theta + \frac{\frac{1}{\sqrt{3}} \cot \theta - 1}{\cot \theta + \frac{1}{\sqrt{3}}} + \frac{-\frac{1}{\sqrt{3}} \cot \theta - 1}{\cot \theta - \frac{1}{\sqrt{3}}}$$

$$= \cot \theta + \frac{\cot \theta - \sqrt{3}}{\sqrt{3} \cot \theta + 1} - \frac{\cot \theta + \sqrt{3}}{\sqrt{3} \cot \theta - 1}$$

$$\cot \theta + \frac{(\cot \theta - \sqrt{3})(\sqrt{3} \cot \theta - 1)}{(3 \cot^2 \theta - 1)} -$$

$$\frac{(\cot \theta + \sqrt{3})(\sqrt{3} \cot \theta + 1)}{(3 \cot^2 \theta - 1)}$$

$$\begin{aligned}
& \cot \theta + \frac{\left[\frac{\sqrt{3} \cot^2 \theta - \cot \theta - 3 \cot \theta + \sqrt{3}}{\sqrt{3} \cot^2 \theta - \cot \theta - 3 \cot \theta - \sqrt{3}} \right]}{(3 \cot^2 \theta - 1)} \\
&= \cot \theta + \left(\frac{-8 \cot \theta}{3 \cot^2 \theta - 1} \right) \\
&= \frac{3 \cot^3 \theta - \cot \theta - 8 \cot \theta}{3 \cot^2 \theta - 1} \\
&= \frac{3 \cot^3 \theta - 9 \cot \theta}{3 \cot^2 \theta - 1} = 3 \left(\frac{\cot^3 \theta - 3 \cot \theta}{3 \cot^2 \theta - 1} \right) = 3 \cot 3\theta
\end{aligned}$$

11. (a); LHS = $\tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + 8 \cot 8\alpha$

$$\begin{aligned}
&= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{8}{\tan 8\alpha} \\
&= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{8}{\tan 2(4\alpha)} \\
&= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{8(1 - \tan^2 4\alpha)}{2 \tan 4\alpha} \\
&= \tan \alpha + 2 \tan 2\alpha + 4 \tan 4\alpha + \frac{4(1 - \tan^2 4\alpha)}{\tan 4\alpha} \\
&= \tan \alpha + 2 \tan 2\alpha + \frac{4 \tan^2 4\alpha + 4 - 4 \tan^2 4\alpha}{\tan 4\alpha} \\
&= \tan \alpha + 2 \tan 2\alpha + \frac{4}{\tan 2(2\alpha)} \\
&= \tan \alpha + 2 \tan 2\alpha + 4 \cdot \frac{1 - \tan^2 2\alpha}{2 \tan 2\alpha} \\
&= \tan \alpha + 2 \tan 2\alpha + 2 \cdot \frac{(1 - \tan^2 2\alpha)}{\tan 2\alpha} \\
&= \tan \alpha + \frac{2}{\tan 2\alpha} = \tan \alpha + \frac{2(1 - \tan^2 \alpha)}{2 \tan \alpha} \\
&= \tan \alpha + \frac{1 - \tan^2 \alpha}{\tan \alpha} = \tan \alpha + \frac{1}{\tan \alpha} - \tan \alpha \\
&= \cot \alpha
\end{aligned}$$

12. (b) $\frac{1}{\cos 290^\circ} + \frac{1}{\sqrt{3} \sin 250^\circ}$

$$\begin{aligned}
&\frac{1}{\sin 20^\circ} - \frac{1}{\sqrt{3} \cos 20^\circ} = \frac{\sqrt{3} \cos 20^\circ - \sin 20^\circ}{\sqrt{3} \sin 20^\circ \cos 20^\circ} \\
&= \frac{2 \cdot \left(\frac{\sqrt{3}}{2} \cos 20^\circ - \frac{1}{2} \sin 20^\circ \right)}{\sqrt{3} \left(\frac{2 \sin 20^\circ \cos 20^\circ}{2} \right)}
\end{aligned}$$

$$= \frac{4(\sin 60^\circ \cos 20^\circ - \cos 60^\circ \sin 20^\circ)}{\sqrt{3} \sin 40^\circ}$$

$$= \frac{4 \sin(60^\circ - 20^\circ)}{\sqrt{3} \sin 40^\circ} = \frac{4}{\sqrt{3}}$$

13. (c) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$

$$\begin{aligned}
&= \tan 9^\circ - \tan 27^\circ - \cot 27^\circ + \cot 9^\circ \\
&= (\tan 9^\circ + \cot 9^\circ) - (\tan 27^\circ + \cot 27^\circ) \\
&= \frac{1}{\sin 9^\circ \cos 9^\circ} - \frac{1}{\sin 27^\circ \cos 27^\circ} \\
&\quad \left(\because \tan \theta + \cot \theta = \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \frac{1}{\sin \theta \cos \theta} \right) \\
&= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} \\
&= 2 \left(\frac{\sin 54^\circ - \sin 18^\circ}{\sin 18^\circ \sin 54^\circ} \right) = 4
\end{aligned}$$

14. (a) $\frac{\sin \alpha + \sin \beta}{\cos \alpha + \cos \beta} = \frac{21}{27}$

$$\begin{aligned}
&\Rightarrow \frac{2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)}{2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right)} = \frac{7}{9} \\
&\Rightarrow \tan \left(\frac{\alpha + \beta}{2} \right) = \frac{7}{9}
\end{aligned}$$

$$\sin \alpha + \sin \beta = -\frac{21}{65}$$

$$\Rightarrow 2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) = -\frac{21}{65}$$

$$2 \cdot \frac{7}{\sqrt{7^2 + 9^2}} \cdot \cos \left(\frac{\alpha - \beta}{2} \right) = -\frac{21}{65}$$

$$\cos \left(\frac{\alpha - \beta}{2} \right) = -\frac{21}{65} \times \frac{\sqrt{130}}{14}$$

$$= \frac{-3 \times \sqrt{130}}{65 \times 2} = \frac{-3}{\sqrt{130}}$$

15. (c) $\cos (79^\circ 53' 28'')$

$$\begin{aligned}
&= \cos (90^\circ - 10^\circ 6' 32'') + \tan (10^\circ 6' 32'') \\
&= \cos (90^\circ - 10^\circ 6' 32'') + \frac{\sin (10^\circ 6' 32'')}{\cos (10^\circ 6' 32'')} \\
&= \sin 10^\circ 6' 32'' + \frac{a}{\sqrt{1 - a^2}} = a + \frac{a}{\sqrt{1 - a^2}}
\end{aligned}$$

$$= \frac{a\sqrt{1-a^2} + a}{\sqrt{1-a^2}} = \frac{a(1+\sqrt{1-a^2})}{\sqrt{1-a^2}}$$

16. (c); $12 \sin \theta = 13 - 5 \cos \theta$
 Or, $144 \sin^2 \theta = 169 + 25 \cos^2 \theta - 130 \cos \theta$
 (squaring both sides)
 Or, $144 (1 - \cos^2 \theta) = 169 + 25 \cos^2 \theta - 130 \cos \theta$
 Or, $144 - 144 \cos^2 \theta = 169 + 25 \cos^2 \theta - 130 \cos \theta$
 Or, $169 \cos^2 \theta - 130 \cos \theta + 25 = 0$
 $(13 \cos \theta - 5)^2 = 0$
 $13 \cos \theta - 5 = 0$

$$\text{Or, } \cos \theta = \frac{5}{13}$$

$$\sin \theta = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$$

$$\operatorname{cosec} \theta = \frac{13}{12}$$

17. (a); $\frac{\sec 29^\circ}{\operatorname{cosec} 61^\circ} = \frac{\sec 29^\circ}{\operatorname{cosec}(90^\circ - 29^\circ)} = \frac{\sec 29^\circ}{\sec 29^\circ} \dots (i)$

$$\begin{aligned} & 2 \cot 8^\circ \cot 17^\circ \cot 45^\circ \cot 73^\circ \cot 82^\circ \\ &= 2 \cot 8^\circ \cdot \cot 17^\circ \cdot 1 \cdot \cot(90^\circ - 17^\circ) \cdot \cot(90^\circ - 8^\circ) \\ & (\because \cot 45^\circ = 1) \\ &= 2 \cot 8^\circ \cot 17^\circ \cdot \tan 17^\circ \cdot \tan 8^\circ \\ &= 2 \cdot 1 \cdot 1 = 2 \end{aligned} \dots (ii)$$

$$\begin{aligned} &= 3(\sin^2 38^\circ + \sin^2 52^\circ) \\ &= 3\{\sin^2 38^\circ + \sin^2(90^\circ - 38^\circ)\} \\ &= 3\{\sin^2 38^\circ + \cos^2 38^\circ\} = 3 \cdot 1 = 3 \end{aligned} \dots (iii)$$

$$\begin{aligned} &= 4(\cos \theta \cos \theta + \sin \theta \sin \theta) \\ &= 4(\cos^2 \theta + \sin^2 \theta) = 4 \end{aligned} \dots (iv)$$

$$\text{From (i), (ii), (iii) (iv)}$$

$$\text{Given expression} = 1 + 2 - 3 + 4 = 4$$

18. (b); LHS = $\sin \theta(1 + \tan \theta) + \cos \theta(1 + \cot \theta)$

$$= \sin \theta \left(1 + \frac{\sin \theta}{\cos \theta} \right) + \cos \theta \left(1 + \frac{\cos \theta}{\sin \theta} \right)$$

$$= \sin \theta \left(\frac{\cos \theta + \sin \theta}{\cos \theta} \right) + \cos \theta \left(\frac{\sin \theta + \cos \theta}{\sin \theta} \right)$$

$$= (\cos \theta + \sin \theta) \left(\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \right)$$

$$= (\cos \theta + \sin \theta) \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} \right)$$

$$= (\cos \theta + \sin \theta) \frac{1}{\cos \theta \sin \theta}$$

$$= \frac{\cos \theta}{\cos \theta \sin \theta} + \frac{\sin \theta}{\cos \theta \sin \theta} = \operatorname{cosec} \theta + \sec \theta$$

19. (a); $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1}$

$$= \frac{\sin \theta + (1 - \cos \theta)}{\sin \theta - (1 - \cos \theta)} \times \frac{\sin \theta + (1 - \cos \theta)}{\sin \theta + (1 - \cos \theta)}$$

[on rationalising]

$$= \frac{\sin^2 \theta + (1 - \cos \theta)^2 + 2 \sin \theta(1 - \cos \theta)}{\sin^2 \theta - (1 - \cos \theta)^2}$$

$$\begin{aligned} & \frac{1 + \sin^2 \theta + \cos^2 \theta - 2 \cos \theta \cdot \sin \theta +}{2 \sin \theta - 2 \cos \theta} \\ &= \frac{\sin^2 \theta - 1 + 2 \cos \theta - \cos^2 \theta}{\sin^2 \theta - 1 + 2 \cos \theta - \cos^2 \theta} \end{aligned}$$

$$= \frac{1 + 1 + 2 \sin \theta - 2 \cos \theta \cdot \sin \theta - 2 \cos \theta}{2 \cos \theta - 1 + 1 - \cos^2 \theta - \cos^2 \theta}$$

$$[2 \sin^2 \theta = 1 - \cos 2\theta]$$

$$= \frac{2(1 - \cos \theta) + 2 \sin \theta(1 - \cos \theta)}{2 \cos \theta(1 - \cos \theta)}$$

$$= \frac{2(1 + \sin \theta)(1 - \cos \theta)}{2 \cos \theta(1 - \cos \theta)} = \frac{1 + \sin \theta}{\cos \theta}$$

20. (c); $\frac{\sec \theta + \tan \theta - 1}{\tan \theta - \sec \theta + 1}$

$$= \frac{(\sec \theta + \tan \theta) - (\sec^2 \theta - \tan^2 \theta)}{\tan \theta - \sec \theta + 1}$$

$$= \frac{(\sec \theta + \tan \theta)(1 - \sec \theta + \tan \theta)}{\tan \theta - \sec \theta + 1}$$

$$= \sec \theta + \tan \theta = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}$$

$$= \frac{1 + \sin \theta}{\cos \theta} \times \frac{1 - \sin \theta}{1 - \sin \theta}$$

$$= \frac{1 - \sin^2 \theta}{\cos \theta(1 - \sin \theta)} = \frac{\cos \theta}{1 - \sin \theta}$$

21. (d); $m^2 - n^2 = (\cot \theta + \cos \theta)^2 - (\cot \theta - \cos \theta)^2$
 $= 4 \cot \theta \cos \theta \dots (i)$

$$= 4\sqrt{mn} = 4\sqrt{(\cot \theta + \cos \theta)(\cot \theta - \cos \theta)}$$

$$= 4\sqrt{\cot^2 \theta - \cos^2 \theta}$$

$$= 4\sqrt{\frac{\cos^2 \theta}{\sin^2 \theta} - \cos^2 \theta} = 4\sqrt{\frac{\cos^2 \theta - \cos^2 \theta \sin^2 \theta}{\sin^2 \theta}}$$

$$= 4\sqrt{\frac{\cos^2 \theta (1 - \sin^2 \theta)}{\sin^2 \theta}} = 4\sqrt{\frac{\cos^2 \theta \cdot \cos^2 \theta}{\sin^2 \theta}}$$

$$= 4\sqrt{\cot^2 \theta \cdot \cos^2 \theta} = 4 \cot \theta \cos \theta \quad \dots (ii)$$

Equation (1) and (2)

$$\frac{m^2 - n^2}{4\sqrt{mm}} = \frac{4 \cot \theta \cos \theta}{4 \cot \theta \cos \theta} = 1$$

22. (a); $\sec^2 \theta - \tan^2 \theta = 1$
 $\therefore (\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1$
 Or, $(\sec \theta - \tan \theta)p = 1$

Or $\sec \theta - \tan \theta = \frac{1}{p} \quad \dots (i)$

$\sec \theta + \tan \theta = p \quad \dots (ii)$

Adding (i) and (ii)

$$2 \sec \theta = \frac{1}{p} + p = \frac{1+p^2}{p}$$

Or, $\sec \theta = \frac{1+p^2}{2p}$

$$\Rightarrow \cos \theta = \frac{2p}{1+p^2}$$

$$= \frac{\text{base}(b)}{\text{hypotenuse}(h)}$$

$$\frac{b}{h} = \frac{2p}{1+p^2}$$

$$p^2 = h^2 - b^2 = (1+p^2)^2 - 4p^2 = (1-p^2)^2$$

$$\sin \theta = \frac{p}{h} = \frac{1-p^2}{1+p^2}$$

23. (b); $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}} - \sec \theta = \sqrt{\frac{1+\sin \theta}{1-\sin \theta} \times \frac{1+\sin \theta}{1+\sin \theta}} - \sec \theta$

$$= \sqrt{\frac{(1+\sin \theta)^2}{1-\sin^2 \theta}} - \sec \theta = \sqrt{\frac{(1+\sin \theta)^2}{\cos^2 \theta}} - \sec \theta$$

$$= \frac{1+\sin \theta}{\cos \theta} - \frac{1}{\cos \theta}, \quad = \frac{1+\sin \theta - 1}{\cos \theta} = \tan \theta$$

$$= \sec \theta - \sqrt{\frac{1-\sin \theta}{1+\sin \theta}} = \sec \theta - \sqrt{\frac{1-\sin \theta}{1+\sin \theta} \times \frac{1-\sin \theta}{1-\sin \theta}}$$

$$= \sec \theta - \sqrt{\frac{(1-\sin \theta)^2}{1-\sin^2 \theta}} = \frac{1}{\cos \theta} - \frac{(1-\sin \theta)}{\cos \theta}$$

$$= \frac{1-1+\sin \theta}{\cos \theta} = \frac{\sin \theta}{\cos \theta} = \tan \theta \quad \dots (2)$$

From (1) and (2)

The given expression = $\frac{\tan \theta}{\tan \theta} = 1$

24. (a); $(1+\sin \alpha)(1+\sin \beta)(1+\sin \gamma)$
 $= (1-\sin \alpha)(1-\sin \beta)(1-\sin \gamma) = x$ let
 $\therefore (1+\sin \alpha)(1+\sin \beta)(1+\sin \gamma) = x \quad \dots (i)$
 and $(1-\sin \alpha)(1-\sin \beta)(1-\sin \gamma) = x \quad \dots (ii)$
 (i) $\times$ (ii)

$$(1-\sin^2 \alpha)(1-\sin^2 \beta)(1-\sin^2 \gamma) = x^2$$

$$\cos^2 \alpha \cos^2 \beta \cos^2 \gamma = x^2$$

$$\therefore x = \pm \cos \alpha \cos \beta \cos \gamma$$

25. (b); $(1+\cot \theta - \operatorname{cosec} \theta)(1+\tan \theta + \sec \theta)$

$$= \left(1 + \frac{\cos \theta}{\sin \theta} - \frac{1}{\sin \theta}\right) \left(1 + \frac{\sin \theta}{\cos \theta} + \frac{1}{\cos \theta}\right)$$

$$= \left(\frac{\sin \theta + \cos \theta - 1}{\sin \theta}\right) \left(\frac{\cos \theta + \sin \theta + 1}{\cos \theta}\right)$$

$$= \frac{(\sin \theta + \cos \theta)^2 - 1^2}{\sin \theta \cdot \cos \theta}, \quad = \frac{2 \sin \theta \cdot \cos \theta}{\sin \theta \cdot \cos \theta} = 2$$

26. (a); $\cos(\alpha - \beta) = \cos[(\alpha - \theta) - (\beta - \theta)]$
 $= \cos(\alpha - \theta) \cos(\beta - \theta) + \sin(\alpha - \theta) \sin(\beta - \theta)$

$$= ab + \sqrt{1-a^2} \cdot \sqrt{1-b^2}$$

$$\left(\because \sin(\alpha - \theta) = \sqrt{1 - \cos^2(\alpha - \theta)}\right)$$

Now, $\sin^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$
 $= 1 - \cos^2(\alpha - \beta) + 2ab \cos(\alpha - \beta)$

$$= 1 - \left(ab + \sqrt{1-a^2} \cdot \sqrt{1-b^2}\right)^2 +$$

$$2ab \cdot \left(ab + \sqrt{1-a^2} \cdot \sqrt{1-b^2}\right)$$

$$= 1 - a^2 b^2 - [1 - b^2 - a^2 + a^2 b^2]$$

$$- 2ab \sqrt{1-a^2} \sqrt{1-b^2} +$$

$$2a^2 b^2 + 2ab \sqrt{1-a^2} \sqrt{1-b^2}$$

$$= 1 - a^2 b^2 - 1 + b^2 + a^2 - a^2 b^2 + 2a^2 b^2$$

$$= a^2 + b^2$$

$$\begin{aligned}
27. \quad (d) \quad & \cos^4 \frac{\pi}{8} + \cos^4 \frac{3\pi}{8} + \cos^4 \frac{5\pi}{8} + \cos^4 \frac{7\pi}{8} \\
&= \cos^4 \frac{\pi}{8} + \cos^4 \left(\frac{\pi}{2} - \frac{\pi}{8} \right) + \cos^4 \left(\frac{\pi}{2} + \frac{\pi}{8} \right) + \cos^4 \left(\pi - \frac{\pi}{8} \right) \\
&= \cos^4 \frac{\pi}{8} + \sin^4 \frac{\pi}{8} + \cos^4 \frac{\pi}{8} + \sin^4 \frac{\pi}{8} \\
&= 2 \left(\cos^4 \frac{\pi}{8} + \sin^4 \frac{\pi}{8} + 2 \cos^2 \frac{\pi}{8} \cdot \sin^2 \frac{\pi}{8} - 2 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right) \\
&= 2 \left[\left(\cos^2 \frac{\pi}{8} + \sin^2 \frac{\pi}{8} \right)^2 - \frac{1}{2} \left(2 \sin^2 \frac{\pi}{8} \cos^2 \frac{\pi}{8} \right)^2 \right] \\
&= 2 \left[(1)^2 - \frac{1}{2} \left(\sin \frac{\pi}{4} \right)^2 \right] = 2 \left[1 - \frac{1}{2} \times \frac{1}{2} \right] = \frac{3}{2}
\end{aligned}$$

$$\begin{aligned}
28. \quad (b) \quad & \sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 4x}}} = \sqrt{2 + \sqrt{2 + \sqrt{2(1 + \cos 4x)}}} \\
&= \sqrt{2 + \sqrt{2 + \sqrt{4 \cos^2 2x}}} = \sqrt{2 + \sqrt{2(1 + \cos 2x)}} \\
&= \sqrt{2 + \sqrt{4 \cos^2 x}} = \sqrt{2 + 2 \cos x} \\
&= \sqrt{2(1 + \cos x)} = 2 \cos \frac{x}{2} \\
&= \frac{2}{\sqrt{2 + \sqrt{2 + \sqrt{2 + 2 \cos 4x}}}} = \frac{2}{2 \cos \frac{x}{2}} = \sec \frac{x}{2}
\end{aligned}$$

$$\begin{aligned}
29. \quad (c); \quad & \left(1 + \cos \frac{\pi}{8} \right) \left(1 + \cos \frac{3\pi}{8} \right) \left(1 + \cos \frac{5\pi}{8} \right) \left(1 + \cos \frac{7\pi}{8} \right) \\
&= \left(2 + \cos^2 \frac{\pi}{16} \right) \cdot \left(2 + \cos^2 \frac{3\pi}{16} \right) \\
&\quad \cdot \left(2 + \cos^2 \frac{5\pi}{16} \right) \cdot \left(2 + \cos^2 \frac{7\pi}{16} \right) \\
&= 16 \left[\left(\cos \frac{\pi}{16} \right) \cdot \left(\cos \frac{7\pi}{16} \right) \left(\cos \frac{3\pi}{16} \right) \cdot \left(\cos \frac{5\pi}{16} \right) \right]^2
\end{aligned}$$

$$\begin{aligned}
&= 16 \left[\frac{1}{2} \left\{ \cos \left(\frac{\pi}{16} + \frac{7\pi}{16} \right) + \cos \left(\frac{7\pi}{16} - \frac{\pi}{16} \right) \right\} \right]^2 \\
&= 16 \left[\frac{1}{2} \left\{ \cos \left(\frac{3\pi}{16} + \frac{5\pi}{16} \right) + \cos \left(\frac{5\pi}{16} - \frac{3\pi}{16} \right) \right\} \right]^2 \\
&= 16 \times \frac{1}{16} \left[\left(\cos \frac{\pi}{2} + \cos \frac{2\pi}{8} \right) \left(\cos \frac{\pi}{2} + \cos \frac{\pi}{8} \right) \right]^2 \\
&= \left[\cos \frac{3\pi}{8} \cdot \cos \frac{\pi}{8} \right]^2 \\
&= \frac{1}{4} \left[\cos \left(\frac{3\pi}{8} + \frac{\pi}{8} \right) + \cos \left(\frac{3\pi}{8} - \frac{\pi}{8} \right) \right]^2 \\
&= \frac{1}{4} \left[\cos \frac{\pi}{2} + \cos \frac{\pi}{4} \right]^2 = \frac{1}{4} \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{8}
\end{aligned}$$

$$\begin{aligned}
30. \quad (d) \quad & (\sec x \cdot \sec y + \tan x \cdot \tan y)^2 - (\sec x \cdot \tan y + \tan x \cdot \sec y)^2 \\
&= \left(\frac{1}{\cos x \cdot \cos y} + \frac{\sin x \sin y}{\cos x \cdot \cos y} \right)^2 - \left(\frac{\sin y}{\cos x \cdot \cos y} + \frac{\sin x}{\cos x \cdot \cos y} \right)^2 \\
&= \left(\frac{1 + \sin x \sin y}{\cos x \cdot \cos y} \right)^2 - \left(\frac{\sin x + \sin y}{\cos x \cdot \cos y} \right)^2 \\
&= \frac{[1 + \sin^2 x \sin^2 y + 2 \sin x \sin y - \sin^2 x - \sin^2 y] - [\sin^2 x + \sin^2 y + 2 \sin x \sin y]}{\cos^2 x \cdot \cos^2 y} \\
&= \frac{1 + \sin^2 x \sin^2 y - \sin^2 x - \sin^2 y}{\cos^2 x \cdot \cos^2 y} \\
&= \frac{\sin^2 x (\sin^2 y - 1) + 1(1 - \sin^2 y)}{\cos^2 x \cdot \cos^2 y} \\
&= 1
\end{aligned}$$

Previous Year (Memory Based)

$$\begin{aligned}
 1. \quad (a) \quad & \frac{\sin A}{1 + \cos A} + \frac{\sin A}{1 - \cos A} \\
 &= \frac{\sin A(1 - \cos A) + \sin A(1 + \cos A)}{1 - \cos^2 A} \\
 &= \frac{\sin A - \sin A \cos A + \sin A + \sin A \cos A}{\sin^2 A} \\
 &= \frac{2 \sin A}{\sin^2 A} = 2 \operatorname{cosec} A
 \end{aligned}$$

$$2. \quad (d) \quad \tan \theta = \frac{r \sin \theta}{r \cos \theta} = \frac{1}{\sqrt{3}}$$

$$\sqrt{3} \tan \theta + 1 = \sqrt{3} \times \frac{1}{\sqrt{3}} + 1 = 2$$

$$\begin{aligned}
 3. \quad (c) \quad & \sin^2 65^\circ + \sin^2 25^\circ + \cos^2 35^\circ + \cos^2 55^\circ \\
 &= \sin^2 65^\circ + \sin^2 (90^\circ - 65^\circ) + \cos^2 35^\circ + \cos^2 (90^\circ - 35^\circ) \\
 &= (\sin^2 65^\circ + \cos^2 65^\circ) + (\cos^2 35^\circ + \sin^2 35^\circ) \\
 &= 1 + 1 = 2
 \end{aligned}$$

$$4. \quad (c) \quad x \sin 60^\circ \cdot \tan 30^\circ = \sec 60^\circ \cdot \cot 45^\circ$$

$$x \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} = 2 \cdot 1, \quad x = 4$$

$$\begin{aligned}
 5. \quad (d) \quad & \frac{1}{2} \sqrt{1 + \sin \theta} + \frac{1}{2} \sqrt{1 - \sin \theta} \\
 &= \frac{1}{2} (\sqrt{1 + \sin 60^\circ} + \sqrt{1 - \sin 60^\circ}) \\
 &= \frac{1}{2} \left(\sqrt{1 + \frac{\sqrt{3}}{2}} + \sqrt{1 - \frac{\sqrt{3}}{2}} \right) \\
 &= \frac{1}{2\sqrt{2}} (\sqrt{2 + \sqrt{3}} + \sqrt{2 - \sqrt{3}}) \\
 &= \frac{1}{2\sqrt{2}} \times \frac{1}{\sqrt{2}} (\sqrt{4 + 2\sqrt{3}} + \sqrt{4 - 2\sqrt{3}}) \\
 &= \frac{1}{4} (\sqrt{(\sqrt{3} + 1)^2} + \sqrt{(\sqrt{3} - 1)^2}) \\
 &= \frac{1}{4} (\sqrt{3} + 1 + \sqrt{3} - 1) \\
 &= \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2} = \cos 30^\circ = \cos \frac{\theta}{2}
 \end{aligned}$$

$$6. \quad (b) \quad \frac{2 \times \left(\frac{1}{\sqrt{3}} \right)^2}{1 - \left(\frac{1}{\sqrt{3}} \right)^2} + (\sqrt{2})^2 - 1 = x \cdot 2$$

$$\frac{\frac{2}{3}}{1 - \frac{1}{3}} + 2 - 1 = 2x, \quad x = 1$$

$$7. \quad (b) \quad \tan \theta = \frac{\sin \alpha - \cos \alpha}{\sin \alpha + \cos \alpha}$$

$$\therefore 1 + \tan^2 \theta = 1 + \frac{(\sin \alpha - \cos \alpha)^2}{(\sin \alpha + \cos \alpha)^2}$$

$$\sec^2 \theta = \frac{(\sin \alpha + \cos \alpha)^2 + (\sin \alpha - \cos \alpha)^2}{(\sin \alpha + \cos \alpha)^2}$$

$$\sec^2 \theta = \frac{2(\sin^2 \alpha + \cos^2 \alpha)}{(\sin \alpha + \cos \alpha)^2}$$

$$\frac{1}{\cos^2 \theta} = \frac{2}{(\sin \alpha + \cos \alpha)^2}$$

$$\frac{1}{\cos \theta} = \frac{\pm \sqrt{2}}{\sin \alpha + \cos \alpha}$$

$$\Rightarrow \sin \alpha + \cos \alpha = \pm \sqrt{2} \cos \theta$$

$$\begin{aligned}
 8. \quad (a); \quad & 4 \sin^2 \theta + 3 \sin^2 \theta + 3 \cos^2 \theta = 4 \\
 & 4 \sin^2 \theta + 3 (\sin^2 \theta + \cos^2 \theta) = 4 \\
 & 4 \sin^2 \theta + 3 = 4, \quad 4 \sin^2 \theta = 1
 \end{aligned}$$

$$\Rightarrow \sin^2 \theta = \frac{1}{4}$$

$$\text{So, } \sin \theta = \frac{1}{2} \quad (0^\circ < \theta < 90^\circ), \quad \theta = 30^\circ$$

$$\cos \theta = \frac{\sqrt{3}}{2}, \quad \tan^2 \theta = \left(\frac{1}{\sqrt{3}} \right)^2 = \frac{1}{3}$$

$$9. \quad (d) \quad \tan 9^\circ = \frac{p}{q} = \cot 81^\circ$$

$$\tan 81^\circ = \frac{1}{\cot 81^\circ} = \frac{q}{p}$$

$$\frac{\sec^2 81^\circ}{1 + \cot^2 81^\circ} = \frac{1 + \tan^2 81^\circ}{1 + \cot^2 81^\circ}$$

$$= \frac{1 + \frac{q^2}{p^2}}{1 + \frac{p^2}{q^2}} = \frac{\frac{p^2 + q^2}{p^2}}{\frac{q^2 + p^2}{q^2}} = \frac{q^2}{p^2}$$

10. (d); $\sec \theta + \tan \theta = 5$... (i)

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$\sec \theta - \tan \theta = \frac{1}{5} \quad \dots (ii)$$

$$\tan \theta = \frac{12}{5}$$

$$\frac{\tan \theta + 1}{\tan \theta - 1} = \frac{\frac{12}{5} + 1}{\frac{12}{5} - 1} = \frac{17}{7}$$

11. (d) $\sec \theta + \tan^3 \theta \operatorname{cosec} \theta = \frac{1}{\cos \theta} + \frac{\sin^3 \theta}{\cos^3 \theta} \cdot \frac{1}{\sin \theta}$

$$= \frac{1}{\cos \theta} + \frac{\sin^2 \theta}{\cos^3 \theta} = \frac{\sin^2 \theta + \cos^2 \theta}{\cos^3 \theta} = \frac{1}{\cos^3 \theta} = \sec^3 \theta$$

$$\tan^2 \theta = 1 - e^2$$

$$\sec^2 \theta = 1 + \tan^2 \theta = 1 + 1 - e^2 = 2 - e^2$$

$$\sec \theta = (2 - e^2)^{\frac{1}{2}} \quad \sec^3 \theta = (2 - e^2)^{\frac{3}{2}}$$

12. (a) $\therefore 2\sin^2 \theta + 3\cos^2 \theta$
 $= 2\sin^2 \theta + 3(1 - \sin^2 \theta)$
 $= 2\sin^2 \theta + 3 - 3\sin^2 \theta$
 $= 3 - \sin^2 \theta$

$$\text{Maximum value of } \sin \theta = 1$$

$$\therefore \text{Minimum value of } 3 - \sin^2 \theta$$

$$= 3 - \sin^2 \theta = 3 - (1)^2 = 2$$

13. (a) $x \cdot \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}} - 1 = \frac{2}{\sqrt{3}} \times \sqrt{3} - 2$

$$\frac{x}{2} - 1 = 0 \quad \frac{x}{2} = 1 \Rightarrow x = 2$$

14. (c) $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2}$
 $= \sec^2 \alpha \cos^2 \beta + \sec^2 \alpha \sin^2 \beta - \tan^2 \alpha$
 $= \sec^2 \alpha [\cos^2 \beta + \sin^2 \beta] - \tan^2 \alpha$
 $= \sec^2 \alpha - \tan^2 \alpha = 1$

15. (c) $\frac{\cos \alpha}{\cos \beta} = a$

$$\cos \alpha = a \cos \beta$$

Squaring both sides

$$\cos^2 \alpha = a^2 \cos^2 \beta$$

$$1 - \sin^2 \alpha = a^2 (1 - \sin^2 \beta) \quad \dots (i)$$

$$\text{Given, } \frac{\sin \alpha}{\sin \beta} = b$$

$$\sin \alpha = b \sin \beta$$

$$\sin^2 \alpha = b^2 \sin^2 \beta \quad \dots (ii)$$

From (i) and (ii)

$$1 - b^2 \sin^2 \beta = a^2 - a^2 \sin^2 \beta$$

$$a^2 \sin^2 \beta - b^2 \sin^2 \beta = a^2 - 1$$

$$\sin^2 \beta = \frac{a^2 - 1}{a^2 - b^2}$$

16. (b) $\frac{\cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

$$= \left(\frac{1}{2}\right)^2 + 4 \times \left(\frac{2}{\sqrt{3}}\right)^2 - 1$$

$$= \frac{3 + 64 - 12}{12} = \frac{55}{12}$$

17. (a) $\sin^2 30^\circ \cos^2 45^\circ + 5 \tan^2 30^\circ + \frac{3}{2} \sin^2 90^\circ$
 $- 3 \cos^2 90^\circ$

$$= \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{\sqrt{2}}\right)^2 + 5 \times \left(\frac{1}{\sqrt{3}}\right)^2 + \frac{3}{2} \times 1 - 0$$

$$= \frac{1}{8} + \frac{5}{3} + \frac{3}{2} = \frac{3 + 40 + 36}{24} = \frac{79}{24} = 3 \frac{7}{24}$$

18. (a) $\cos^4 \theta - \sin^4 \theta$

$$= (\cos^2 \theta + \sin^2 \theta)(\cos^2 \theta - \sin^2 \theta) = 1 \cdot \frac{1}{3} = \frac{1}{3}$$

19. (c) $\sec^2 \theta = 1 + \tan^2 \theta = 1 + \frac{1}{11} = \frac{12}{11}$

$$\cot \theta = \sqrt{11}$$

$$\operatorname{cosec}^2 \theta = 1 + \cot^2 \theta = 1 + 11 = 12$$

$$\frac{\operatorname{cosec}^2 \theta - \sec^2 \theta}{\operatorname{cosec}^2 \theta + \sec^2 \theta} = \frac{12 - \frac{12}{11}}{12 + \frac{12}{11}} = \frac{120}{144} = \frac{5}{6}$$

20. (a) $\frac{1}{\sqrt{2}} \times \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}} \times \frac{2}{\sqrt{3}} + \frac{5}{12}$

$$= \frac{1}{4} - \frac{2}{3} + \frac{5}{12} = \frac{3 - 8 + 5}{12} = 0$$

$$21. (b) \cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{9}{25}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$$

$$\tan\theta = \frac{3/4}{4/5} = \frac{3}{4}, \quad \cot\theta = \frac{4}{3}$$

$$\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{5}{3}$$

$$\frac{\tan\theta + \cos\theta}{\cot\theta + \operatorname{cosec}\theta} = \frac{\frac{3}{4} + \frac{4}{5}}{\frac{4}{3} + \frac{5}{3}} = \frac{15 + 16}{20 \times 3} = \frac{31}{60}$$

$$\begin{aligned} 22. (b); & a \cos\theta + b \sin\theta = p \\ & p^2 = a^2 \cos^2\theta + b^2 \sin^2\theta + 2ab \sin\theta \cos\theta \\ & q = a \sin\theta - b \cos\theta \\ & q^2 = a^2 \sin^2\theta + b^2 \cos^2\theta - 2ab \sin\theta \cos\theta \\ & p^2 + q^2 = a^2 \cos^2\theta + b^2 \sin^2\theta + 2ab \sin\theta \cos\theta + a^2 \sin^2\theta + b^2 \cos^2\theta - 2ab \sin\theta \cos\theta \\ & = a^2(\cos^2\theta + \sin^2\theta) + b^2(\sin^2\theta + \cos^2\theta) \\ & = a^2 + b^2 \\ & \text{so, } a^2 + b^2 = p^2 + q^2 \end{aligned}$$

$$23. (d); \sin\theta + \cos\theta = \frac{7}{5}$$

squaring on both side

$$\sin^2\theta + \cos^2\theta + 2 \sin\theta \cos\theta = \frac{49}{25}$$

$$2 \sin\theta \cos\theta = \frac{49}{25} - 1 = \frac{24}{25}, \quad \sin\theta \cos\theta = \frac{12}{25}$$

$$24. (a) \sin 21^\circ = \frac{x}{y}$$

$$\cos 21^\circ = \sqrt{1 - \frac{x^2}{y^2}} = \sqrt{\frac{y^2 - x^2}{y^2}} = \frac{\sqrt{y^2 - x^2}}{y}$$

$$\sec 21^\circ = \frac{y}{\sqrt{y^2 - x^2}}$$

$$\sin 69^\circ = \sin (90^\circ - 21^\circ) = \cos 21^\circ = \frac{\sqrt{y^2 - x^2}}{y}$$

$$\sec 21^\circ - \sin 69^\circ = \frac{y}{\sqrt{y^2 - x^2}} - \frac{\sqrt{y^2 - x^2}}{y}$$

$$= \frac{y^2 - (y^2 - x^2)}{y\sqrt{y^2 - x^2}} = \frac{x^2}{y\sqrt{y^2 - x^2}}$$

$$25. (c); \sec\alpha - \tan\alpha = \frac{1}{2}, \quad 2\sec\alpha = 2 + \frac{1}{2}$$

$$\sec\alpha = \frac{5}{4}, \quad \cos\alpha = \frac{4}{5}$$

$$\sin\alpha = \sqrt{1 - \cos^2\alpha} = \sqrt{1 - \frac{16}{25}}, \quad \sin\alpha = \frac{3}{5} = 0.6$$

$$26. (a); 3 \sin\theta + 5 \cos\theta = 5$$

squaring both sides :

$$9 \sin^2\theta + 25 \cos^2\theta + 30 \sin\theta \cos\theta = 25 \quad \dots (i)$$

$$\text{Let } 5 \sin\theta - 3 \cos\theta = x$$

$$25 \sin^2\theta + 9 \cos^2\theta - 30 \sin\theta \cos\theta = x^2 \quad \dots (ii)$$

Adding (1) and (2) :

$$34 \sin^2\theta + 34 \cos^2\theta = x^2 + 25$$

$$34 = x^2 + 25, \quad x^2 = 9, \quad x = \pm 3$$

$$27. (b) \tan\theta + \cot\theta = 2$$

$$\tan\theta + \frac{1}{\tan\theta} = 2, \quad 1 + \tan^2\theta = 2 \tan\theta$$

$$1 + \tan^2\theta - 2 \tan\theta = 0, \quad (1 - \tan\theta)^2 = 0$$

$$1 - \tan\theta = 0, \quad \tan\theta = 1$$

$$\cot\theta = \frac{1}{\tan\theta} = 1$$

$$\text{so, } \tan^5\theta + \cot^5\theta = 1 + 1 = 2$$

$$\begin{aligned} 28. (c); & \tan 1^\circ \times \tan 2^\circ \times \tan 3^\circ \dots \tan 89^\circ \\ & = \tan 1^\circ \times \tan 2^\circ \tan 3^\circ \dots \tan 45^\circ \dots \tan 87^\circ \cdot \tan 88^\circ \times \tan 89^\circ \\ & = (\tan 1^\circ \cdot \tan 89^\circ) \cdot (\tan 2^\circ \cdot \tan 88^\circ) \dots \tan 45^\circ \\ & = [\tan 1^\circ \cdot \tan (90^\circ - 1^\circ)] \cdot [\tan 2^\circ \cdot \tan (90^\circ - 2^\circ)] \dots \tan 45^\circ \\ & = (\tan 1^\circ \cdot \cot 1^\circ) \cdot (\tan 2^\circ \cdot \cot 2^\circ) \dots \tan 45^\circ \\ & = 1 \cdot 1 \cdot 1 \cdot 1 \dots 1 = 1 \end{aligned}$$

$$29. (c) x \sin^2 60^\circ - \frac{3}{2} \sec 60^\circ \tan^2 30^\circ + \frac{4}{5} \sin^2 45^\circ \tan^2 60^\circ = 0$$

$$x \times \left(\frac{\sqrt{3}}{2}\right)^2 - \frac{3}{2} \times 2 \times \left(\frac{1}{\sqrt{3}}\right)^2 + \frac{4}{5} \times \left(\frac{1}{\sqrt{2}}\right)^2 \cdot (\sqrt{3})^2 = 0$$

$$\frac{3}{4}x - 3 \times \frac{1}{3} + \frac{4}{5} \times \frac{1}{2} \times 3 = 0$$

$$\frac{3}{4}x - 1 + \frac{6}{5} = 0, \quad \frac{15x - 20 + 24}{20} = 0$$

$$15x + 4 = 0, \quad x = -\frac{4}{15}$$

$$30. (d); \tan\alpha = \frac{24}{7}$$

$$\sec^2\alpha = 1 + \tan^2\alpha = 1 + \left(\frac{24}{7}\right)^2$$

$$= 1 + \frac{576}{49}, \quad = \frac{49 + 576}{49}, \quad = \frac{625}{49}$$

$$\sec^2\alpha = \frac{625}{49}$$

$$\sec\alpha = \sqrt{\frac{625}{49}} = \frac{25}{7} \quad \left[\because 0 < \alpha < \frac{\pi}{2} \right]$$

$$\cos \alpha = \frac{1}{\sec \alpha} = \frac{7}{25}$$

$$14 \tan \alpha - 75 \cos \alpha - 7 \sec \alpha$$

$$= 14 \times \frac{24}{7} - 75 \times \frac{7}{25} - 7 \times \frac{25}{7} = 48 - 21 - 25 = 2$$

31. (b) $2 \operatorname{cosec}^2 30^\circ + x \sin^2 60^\circ - \frac{3}{4} \tan^2 30^\circ = 10$

$$2 \times 2^2 + x \times \left(\frac{\sqrt{3}}{2}\right)^2 - \frac{3}{4} \times \left(\frac{1}{\sqrt{3}}\right)^2 = 10$$

$$8 + \frac{3}{4}x - \frac{1}{4} = 10, \quad 32 + 3x - 1 = 40$$

$$3x = 9, \quad x = 3$$

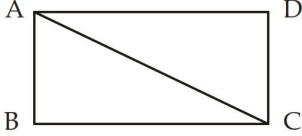
32. (b); $(\sec^2 \theta - \tan^2 \theta) = 1$,
so, $(\tan^2 \theta - \sec^2 \theta) = -1$

33. (b); $29 \tan \theta = 31, \quad \tan \theta = \frac{31}{29}$

$$\frac{1 + 2 \sin \theta \cos \theta}{1 - 2 \sin \theta \cos \theta} = \frac{\sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta}{\sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta}$$

$$= \frac{(\sin \theta + \cos \theta)^2}{(\sin \theta - \cos \theta)^2} = \frac{\left(\frac{\sin \theta}{\cos \theta} + 1\right)^2}{\left(\frac{\sin \theta}{\cos \theta} - 1\right)^2}$$

$$= \left(\frac{\tan \theta + 1}{\tan \theta - 1}\right)^2 = \left(\frac{\frac{31}{29} + 1}{\frac{31}{29} - 1}\right)^2 = \left(\frac{60}{2}\right)^2 = 30^2 = 900$$

34. (c) 

$$\angle CAD = 45^\circ \quad \text{since ABCD is a square}$$

$$\angle BAC = 45^\circ$$

$$(\tan^2 \angle CAD + 1) \sin^2 \angle BAC$$

$$= (\tan^2 45^\circ + 1) \sin^2 45^\circ = (1 + 1) \left(\frac{1}{\sqrt{2}}\right)^2 = 2 \times \frac{1}{2} = 1$$

35. (b) $\tan x = \sin 45^\circ \cdot \cos 45^\circ + \sin 30^\circ$

$$\tan x = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + \frac{1}{2} = 1 = \tan 45^\circ$$

$$\tan x = \tan 45^\circ, \quad x = 45^\circ$$

36. (c); $\sqrt{\frac{\sec \theta - 1}{\sec \theta + 1}} = \sqrt{\frac{\frac{1}{\cos \theta} - 1}{\frac{1}{\cos \theta} + 1}} = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$

$$= \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta} \times \frac{1 - \cos \theta}{1 - \cos \theta}} = \sqrt{\frac{(1 - \cos \theta)^2}{(1 - \cos^2 \theta)}}$$

$$= \sqrt{\frac{(1 - \cos \theta)^2}{\sin^2 \theta}} = \frac{1 - \cos \theta}{\sin \theta} = \frac{1}{\sin \theta} - \frac{\cos \theta}{\sin \theta}$$

$$= \operatorname{cosec} \theta - \cot \theta$$

37. (b) Let the two angles be A and B.

$$A + B = 135^\circ \quad \dots(i)$$

$$A - B = \frac{\pi}{12} = \frac{\pi}{12} \times \frac{180^\circ}{\pi}$$

$$A - B = 15^\circ \quad \dots(ii)$$

Adding (i) and (ii)

$$2A = 150^\circ$$

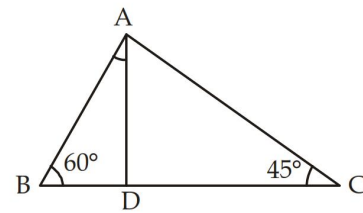
$$A = 75^\circ \quad \dots(iii)$$

From (i) and (iii)

$$B = 60^\circ$$

So, $A = 75^\circ, B = 60^\circ$

38. (c)



$$\frac{BD}{DC} = \frac{1}{3}$$

From $\triangle ABD$:

$$\frac{BD}{\sin \angle BAD} = \frac{AD}{\sin \angle ABD} \quad (\text{By sin rule})$$

$$\frac{BD}{\sin \angle BAD} = \frac{AD}{\sin \frac{\pi}{3}} = \frac{AD}{\frac{\sqrt{3}}{2}}$$

$$\Rightarrow AD = \frac{\sqrt{3}}{2} \cdot \frac{BD}{\sin \angle BAD} \quad \dots(i)$$

from $\triangle ACD$:

$$\frac{CD}{\sin \angle CAD} = \frac{AD}{\sin 45^\circ}$$

$$\frac{CD}{\sin \angle CAD} = \frac{AD}{\frac{1}{\sqrt{2}}}$$

$$AD = \frac{1}{\sqrt{2}} \frac{CD}{\sin \angle CAD} \quad \dots(\text{iii})$$

From (i) and (ii) :

$$\frac{\sqrt{3}}{2} \cdot \frac{BD}{\sin \angle BAD} = \frac{1}{\sqrt{2}} \cdot \frac{CD}{\sin \angle CAD}$$

$$\frac{\sin \angle BAD}{\sin \angle CAD} = \frac{\sqrt{3}}{2} \times \sqrt{2} \cdot \frac{BD}{CD} = \frac{\sqrt{3}}{2} \times \sqrt{2} \times \frac{1}{3}$$

$$\Rightarrow \frac{\sin \angle BAD}{\sin \angle CAD} = \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{6}}$$

$$39. \quad (a) \quad \cos (90^\circ - 3A) = \cos (A - 26^\circ) \\ 90^\circ - 3A = A - 26^\circ$$

$$4A = 116^\circ$$

$$A = 29^\circ$$

$$\begin{aligned} 40. \quad (a) \quad & \sec^2 \theta - \frac{\sin^2 \theta - 2 \sin^4 \theta}{2 \cos^4 \theta - \cos^2 \theta} \\ &= \sec^2 \theta - \frac{\sin^2 (1 - 2 \sin^2 \theta)}{\cos^2 (2 \cos^2 \theta - 1)} \\ &= \sec^2 \theta - \tan^2 \theta \cdot \left(\frac{\cos 2\theta}{\cos 2\theta} \right) \\ &= \sec^2 \theta - \tan^2 \theta \times 1 \\ &= \sec^2 \theta - \tan^2 \theta = 1 \end{aligned}$$